

Real Analysis

An Undergraduate Notebook

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Part I

Analytical Foundations

Chapter 1

Introductory Set Theory

In this chapter, we lay the groundwork for our exposition of real analysis. We will introduce the fundamental notions of sets, relations, and functions, which underlie not only all of analysis but also all of modern mathematics as well.

1.1 What Is a Set?

Since the early 20th century, the practice of mathematics has undergone a radical transformation, which grew out of the idea that the whole of the mathematical canon can and should be rephrased in the language of set theory. Informally, a **set** is a collection of objects, which are called its **elements**. If an object x is an element of a set S , then we write $x \in S$; otherwise, if x is not an element of S , then we write $x \notin S$.

1.1.1 Definitions of Sets

You may complain that the terms “collection” and “object” are vague and undefined, and you would be right to object; the above characterization of sets is definitely not as rigorous as we might desire, and is in fact internally inconsistent.

However, the technical difficulties associated with building a rigorous and internally consistent set theory are outside the scope of this course and in this context, far more trouble than they are worth. In fact, it’s entirely possible that such an internally consistent set theory does not exist!

As such, we will leave this ambiguity in favor of moving on to more salient topics. Even though we will not encounter any such issues in this text, the “definition” above does lead to some interesting inconsistencies. We will explore one such contradiction, called [Russell’s paradox](#), shortly. First, however, we introduce some shorthand notation which will be useful in specifying sets we wish to discuss.

Convention 1.1.1 Roster notation. It’s easy to completely describe a set which contains only a finite number of elements; we can simply write those elements down! Concretely, if S is the set whose elements are precisely the objects $x_1, \dots, x_n$, then we may write

$$S = \{x_1, \dots, x_n\}.$$

The above notation for specifying sets is called **roster notation**. Not all sets are finite, however, and it is obviously impossible to write down a complete

roster of all of the elements of an infinite set. Therefore, we cannot generalize roster notation to sets which contain infinitely many elements.

Definition 1.1.2 Definable collection. Let Φ be a **unary Boolean predicate** (a statement whose truth value depends on a single variable x). The collection of objects x which satisfy Φ is said to be **defined** by Φ , and is denoted

$$\{x \mid \Phi(x)\}.$$

Such a collection of objects which satisfy such a Boolean predicate is called a **definable collection**. We will see that the collection of objects that satisfy a given Boolean predicate is not necessarily a set; this is the content of Russell's paradox. $\blacklozenge$

Example 1.1.3 Sets. The following are examples of sets:

- The **empty set**, denoted $\emptyset$ or $\{\}$, is the set with no elements; that is, $x \notin \emptyset$ for all objects x .
- The set $\{1, 2, \text{red}, \text{blue}\}$ is a **non-empty** set with only finitely many elements. Note that this set contains different types of objects. In general, we will not place any explicit restrictions on the types of objects allowed inside a set, although some formalizations of axiomatic set theories rely on this to avoid inconsistencies such as Russell's paradox.
- The set $\mathbb{N} = \{0, 1, 2, \dots\}$ of **natural numbers** (also sometimes called the **whole numbers** or **counting numbers**) is a non-empty set with infinitely many elements. We will characterize the natural numbers $\mathbb{N}$ in the usual manner of Peano arithmetic later in this chapter.

$\blacktriangle$

As we mentioned before, the set theory based on the “definition” above, which is called **naïve set theory**, is in fact inconsistent; taking this “definition” as the basis for an axiomatic system, we can derive a contradiction, which implies that we may in fact derive all possible statements, including those which are false! This inconsistency was first demonstrated by Bertrand Russell at the beginning of the 20th century, and bears his name.

Proposition 1.1.4 Russell's paradox. *Not all definable collections are sets.*

Proof. By contradiction. Let

$$\mathfrak{R} = \{S \mid S \notin S\}$$

be the definable collection of sets which do not contain themselves, and suppose for a contradiction that $\mathfrak{R}$ is a set. The question of whether or not $\mathfrak{R}$ contains itself is a natural one; however, we will see that both the positive and negative answers lead to contradictions.

This is because, if $\mathfrak{R} \in \mathfrak{R}$, then $\mathfrak{R} \notin \mathfrak{R}$ by the definition of $\mathfrak{R}$. On the other hand, if $\mathfrak{R} \notin \mathfrak{R}$, then $\mathfrak{R} \in \mathfrak{R}$ by the same reasoning. Therefore, $\mathfrak{R} \in \mathfrak{R}$ if and only if $\mathfrak{R} \notin \mathfrak{R}$. This contradiction implies that $\mathfrak{R}$ is not a set, as desired. $\blacksquare$

Russell's paradox tells us that naïve set theory is inconsistent. However, as we stated before, it is no easy task to construct a consistent set theory which is strong enough to describe the sophisticated structures we wish to investigate while also weak enough to be internally consistent. However, since the question of whether modern formal set theories are consistent is beyond the scope of this text. With this in mind, we will continue formulating definitions and results

in the language of naïve set theory, keeping in mind that rigorous systems of formal set theory (which do not suffer from Russell's paradox) exist in which they can be expressed and proven.

1.1.2 Set Containment

As we have just seen, not all definable collections of mathematical objects are sets. However, any collection of objects which themselves belong to a set is also a set in its own right; such a set of elements of a given set is called a subset.

Definition 1.1.5 Subset; superset. A set S is a **subset** of a set T if $x \in T$ for all elements $x \in S$. In this case, we say that S is **contained** in T , and we write $S \subseteq T$. Less commonly, we may call T a **superset** of S if S is a subset of T . In this case, we say that T **contains** S , and we write $T \supseteq S$.

Also, a set S is a **proper subset** of a set T if $S \subseteq T$ but $T \not\subseteq S$. In this case, we say that S is **strictly contained** in T , and we write $S \subsetneq T$. Analogously, T is a **proper superset** of S if S is a proper subset of T . In this case, we say that T **strictly contains** S , and we write $T \supsetneq S$. ♦

Definition 1.1.6 Set equality. Two sets are considered **equal** if they contain exactly the same elements; that is, one set A is equal to another set B when objects are elements of A if and only if they are elements of B . This occurs precisely when both $A \subseteq B$ and $B \subseteq A$. ♦

Example 1.1.7 Examples of subsets. The following are examples of subsets of sets:

- For any set S , the empty set $\emptyset$ and S itself are subsets of S .
- The set $\{1, 3, 5, \dots\}$ of odd natural numbers is a proper subset of the set $\mathbb{N}$ of natural numbers. Written more concisely,

$$\{1, 3, 5, \dots\} \subsetneq \mathbb{N}.$$

▲

Lemma 1.1.8 Transitivity of set containment. *For any sets S , T , and U , if $S \subseteq T$ and $T \subseteq U$, then $S \subseteq U$.*

Proof. Suppose that $S \subseteq T$ and $T \subseteq U$, and let $x \in S$. Since $x \in S$ and $S \subseteq T$, $x \in T$. Moreover, since $x \in T$ and $T \subseteq U$, $x \in U$.

In summary, we have shown that $x \in U$ for all $x \in S$. We conclude that $S \subseteq U$. ■

Convention 1.1.9 Set-builder notation. Although not all definable collections are sets, any definable subcollection of a set is itself a set; that is, given a set S and a unary Boolean predicate Φ , the definable collection

$$\{x \mid x \in S \text{ and } \Phi(x)\}$$

is a set (and in fact a subset of S). This subset will also be denoted in **set-builder notation** by

$$\{x \in S \mid \Phi(x)\} = \{x \mid x \in S \text{ and } \Phi(x)\}.$$

Definition 1.1.10 Power set. The **power set** of a set S is the set of subsets of S ; this set is denoted $\mathcal{P}(S)$, so that

$$\mathcal{P}(S) = \{T \mid T \subseteq S\}.$$



Lemma 1.1.11 Power sets are non-empty. *The power set $\mathcal{P}(S)$ of any set S is non-empty; that is, $\mathcal{P}(S) \neq \emptyset$.*

Proof. Fix a set S , and note that the statement $x \in S$ for all $x \in \emptyset$ is vacuously true, so that $\emptyset \subseteq S$. Thus $\emptyset \in \mathcal{P}(S)$, and so $\mathcal{P}(S) \neq \emptyset$ is non-empty. ■

Proof. Fix a set S , and note that the statement $x \in S$ for all $x \in S$ is a tautology, so that $S \subseteq S$. Thus $S \in \mathcal{P}(S)$, and so $\mathcal{P}(S) \neq \emptyset$ is non-empty. ■

Insight 1.1.12 Analysis of the proofs of Lemma 1.1.11. Both of the above proofs of [Lemma 1.1.11 Power sets are non-empty](#) are correct, although they are nonequivalent. Both are valuable for different reasons; the first proof shows that the power set of a set is nonempty since the empty set $\emptyset$ is a subset of any set, and the second proof shows the same conclusion since any set is a subset of itself. These proofs are called **constructive**, in that they explicitly exhibited an object satisfied the desired properties (being an element of the power set of a given set). Proofs which are not constructive are called **non-constructive**.

In general, sets are likely to have many subsets, so that power sets of sets are likely to have a great many elements. We will investigate the notion of the number of elements of a set, called its **cardinality**, more precisely and in far greater depth at the end of this chapter.

1.1.3 Set Operations

Test reference [Subsection 1.1.3 Set Operations](#)

We now introduce several ways to construct new sets from old ones which loosely correspond to the logical operators of **conjunction** $\wedge$, **disjunction** $\vee$, and **negation** $\neg$, pronounced “and”, “or” and “not”, respectively.

Definition 1.1.13 Intersection; union; complement. The **intersection** $S \cap T$ of two sets S and T is the set whose elements are both elements of S and elements of T ; that is,

$$S \cap T = \{x \mid x \in S \text{ and } x \in T\}.$$

The **union** $S \cup T$ of S and T is the set whose elements are either elements of S or elements of T ; that is,

$$S \cup T = \{x \mid x \in S \text{ or } x \in T\}.$$

The **complement** $S \setminus T$ of T in S is the set whose elements are elements of S but not elements of T ; that is,

$$S \setminus T = \{x \mid x \in S \text{ and } x \notin T\}.$$



Proposition 1.1.14 Set operations and containment. *Fix sets S , T , and U .*

1. $U \subseteq S \cap T$ if and only if $U \subseteq S$ and $U \subseteq T$.
2. $S \cup T \subseteq U$ if and only if $S \subseteq U$ and $T \subseteq U$.
3. $U \subseteq S \setminus T$ if and only if $U \subseteq S$ and $U \cap T = \emptyset$.

Proof.

1. First suppose that $U \subseteq S \cap T$. Since $S \cap T \subseteq S$, [Lemma 1.1.8 Transitivity of set containment](#) implies that $U \subseteq S$. Similarly, since $S \cap T \subseteq T$, [Lemma 1.1.8 Transitivity of set containment](#) implies that $U \subseteq T$.

Conversely, now suppose that $U \subseteq S$ and $U \subseteq T$, and let $x \in U$. Since $x \in U$ and $U \subseteq S$, $x \in S$. Similarly, since $x \in U$ and $U \subseteq T$, $x \in T$. Since $x \in S$ and $x \in T$, we conclude that $x \in S \cap T$. In summary, we have shown that $x \in S \cap T$ for all $x \in U$, so that $U \subseteq S \cap T$.

2. First suppose that $S \cup T \subseteq U$. Since $S \subseteq S \cup T$, [Lemma 1.1.8 Transitivity of set containment](#) implies that $S \subseteq U$. Similarly, since $T \subseteq S \cup T$, [Lemma 1.1.8 Transitivity of set containment](#) implies that $T \subseteq U$.

Conversely, now suppose that $S \subseteq U$ and $T \subseteq U$. Let $x \in S \cup T$, so that $x \in S$ or $x \in T$. If $x \in S$, then $x \in U$, since $S \subseteq U$. Similarly, if $x \in T$, then $x \in U$, since $T \subseteq U$. We conclude that $x \in U$. In summary, we have shown that $x \in U$ for all $x \in S \cup T$, so that $S \cup T \subseteq U$.

3. First suppose that $U \subseteq S \setminus T$, and let $x \in U$. Since $x \in U$ and $U \subseteq S \setminus T$, $x \in S \setminus T$, so that $x \in S$ and $x \notin T$. In summary, we have shown that $x \in S$ for all $x \in U$, so that $U \subseteq S$. Moreover, we have shown that $x \notin T$ for all $x \in U$, so that no element of U is an element of T ; that is, $U \cap T = \emptyset$.

Conversely, now suppose that $U \subseteq S$ and $U \cap T = \emptyset$, and let $x \in U$. Since $x \in U$ and $U \subseteq S$, $x \in S$. Moreover, since $x \in U$ and $U \cap T = \emptyset$, $x \notin T$. Since $x \in S$ and $x \notin T$, we conclude that $x \in S \setminus T$. In summary, we have shown that $x \in S \setminus T$ for all $x \in U$, and so $U \subseteq S \setminus T$.

■

Corollary 1.1.15 Properties of the set operations. *Fix sets S , T , and U .*

1. $S \cap S = S \cup S = S \cup \emptyset = S \setminus \emptyset = S$, and $S \cap \emptyset = S \setminus S = \emptyset \setminus S = \emptyset$.
2. **Commutativity:** $S \cap T = T \cap S$ and $S \cup T = T \cup S$.
3. **Associativity:** $(S \cap T) \cap U = S \cap (T \cap U)$ and $(S \cup T) \cup U = S \cup (T \cup U)$.

Proof.

1. Since $S \subseteq S$, $S \subseteq S \cap S$ and $S \cup S \subseteq S$ by (1) and (2) of [Proposition 1.1.14 Set operations and containment](#), respectively. Of course $S \cap S \subseteq S \subseteq S \cup S$, so that $S \cap S = S = S \cup S$.

Similarly, since $S \subseteq S$ and $\emptyset \subseteq S$, $S \cup \emptyset \subseteq S$ by (2) of [Proposition 1.1.14 Set operations and containment](#). Of course $S \subseteq S \cup \emptyset$, and so $S \cup \emptyset = S$.

$\emptyset \subseteq S \cap \emptyset \subseteq \emptyset$, and so $S \cap \emptyset = \emptyset$. Moreover, together with the observation that $S \subseteq S$, this implies by (1) of [Proposition 1.1.14 Set operations and containment](#) that $S \subseteq S \setminus \emptyset$. Of course, $S \setminus \emptyset \subseteq S$, and so $S \setminus \emptyset = S$.

Similarly, $\emptyset \subseteq \emptyset \setminus S \subseteq \emptyset$, and so $\emptyset \setminus S = \emptyset$.

Finally, note that any element of $S \setminus S$ is both an element of S and not an element of S , so that no such element can exist. Thus $S \setminus S = \emptyset$.

2. Since $S \cap T \subseteq T$ and $S \cap T \subseteq S$, (1) of [Proposition 1.1.14 Set operations and containment](#) implies that $S \cap T \subseteq T \cap S$. By symmetry, $T \cap S \subseteq S \cap T$, so that $S \cap T = T \cap S$.

Similarly, since $S \subseteq T \cup S$ and $T \subseteq T \cup S$, (2) of [Proposition 1.1.14 Set operations and containment](#) implies that $S \cup T \subseteq T \cup S$. By symmetry, $T \cup S \subseteq S \cup T$, so that $S \cup T = T \cup S$.

3. We note that

$$\begin{aligned}(S \cap T) \cap U &\subseteq S \cap T \subseteq T \\ (S \cap T) \cap U &\subseteq U,\end{aligned}$$

so that $(S \cap T) \cap U \subseteq T \cap U$ by (1) of [Proposition 1.1.14 Set operations and containment](#). Similarly,

$$(S \cap T) \cap U \subseteq S \cap T \subseteq T,$$

so that $(S \cap T) \cap U \subseteq S \cap (T \cap U)$ by another application of (1) of [Proposition 1.1.14 Set operations and containment](#).

Conversely, we note that

$$\begin{aligned}S \cap (T \cap U) &\subseteq S \\ S \cap (T \cap U) &\subseteq T \cap U \subseteq T,\end{aligned}$$

so that $S \cap (T \cap U) \subseteq S \cap T$ by (1) of [Proposition 1.1.14 Set operations and containment](#). Similarly,

$$S \cap (T \cap U) \subseteq T \cap U \subseteq U,$$

so that $S \cap (T \cap U) \subseteq (S \cap T) \cap U$ by another application of (1) of [Proposition 1.1.14 Set operations and containment](#). Thus

$$(S \cap T) \cap U = S \cap (T \cap U).$$

■

In the next section, we will generalize the first two ways (**intersection** and **union**) to arbitrary sets of sets; this generalization will mirror the similarities between conjunction $\wedge$ and universal quantification $\forall$ and between disjunction $\vee$ and existential quantification $\exists$.

While sets and set theory form the foundations on which we will build our understanding of analysis, these topics are not the focus of this text. As such, we will leave behind the question of precise definitions of sets in favor of developing set-theoretic ideas which are more relevant to the world of analysis.

1.1.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

1. Power sets. Compute the power set $\mathcal{P}(S)$ of the set $S = \{1, 2, 3\}$.

Answer. The power set $\mathcal{P}(S)$ of the set $S = \{1, 2, 3\}$ is the set

$$\mathcal{P}(S) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

Solution. Choosing a subset of the set $S = \{1, 2, 3\}$ is equivalent to making a unique sequence of three binary choices of whether to include each element 1, 2, and 3. We see that S has the following 8 subsets: $\emptyset$, $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$, and $\{1, 2, 3\}$. Therefore, the power set $\mathcal{P}(S)$ of the set $S = \{1, 2, 3\}$ is the set

$$\mathcal{P}(S) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

- 2. Set Operations.** Compute the given set operations. Write your answer in roster or set-builder notation, or as a well-known set.

- (a) Compute $S \cap T$, where

$$S = \{n \in \mathbb{N} \mid n = 2k \text{ for some } k \in \mathbb{N}\} = \{0, 2, 4, 6, \dots\}$$

is the set of even natural numbers and

$$T = \{n \in \mathbb{N} \mid n = 2k + 1 \text{ for some } k \in \mathbb{N}\} = \{1, 3, 5, 7, \dots\}$$

is the set of odd integers.

Answer. $S \cap T = \emptyset$.

Solution. No natural number is both even and odd, and so $S \cap T = \emptyset$.

- (b) Compute $S \cup T$, where $S = \{0, 3, 4, 9\}$ and $T = \{1, 3, 6\}$.

Answer. $S \cup T = \{0, 1, 3, 4, 6, 9\}$.

Solution. $S \cup T$ contains all the elements of S and all the elements of T , so that $S \cup T = \{0, 1, 3, 4, 6, 9\}$.

- (c) Compute $S \setminus T$, where

$$S = \{n \in \mathbb{N} \mid n = 5k \text{ for some } k \in \mathbb{N}\} = \{0, 5, 10, 15, \dots\}$$

is the set of natural numbers divisible by 5, and

$$T = \{n \in \mathbb{N} \mid n = 2k \text{ for some } k \in \mathbb{N}\} = \{0, 2, 4, 6, \dots\}$$

is the set of even natural numbers.

Answer. $S \setminus T = \{n \in \mathbb{N} \mid n = 10k + 5 \text{ for some } k \in \mathbb{N}\} = \{5, 15, 25, 35, \dots\}$.

Solution. $S \setminus T$ contains all the odd multiples of 5, so that

$$\begin{aligned} S \setminus T &= \{n \in \mathbb{N} \mid n = 10k + 5 \text{ for some } k \in \mathbb{N}\} \\ &= \{5, 15, 25, 35, \dots\}. \end{aligned}$$

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.1.5 Reading Questions

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **De Morgan's laws.** Let S , T , and U be sets.

(a) Prove that $S \setminus (T \cap U) = (S \setminus T) \cup (S \setminus U)$.

Solution. asdf

(b) Prove that $S \setminus (T \cup U) = (S \setminus T) \cap (S \setminus U)$.

Solution. asdf

Taken together, these results are called **De Morgan's laws**.

2. **The distributive properties of set operations.** Let S , T , and U be sets.

(a) Prove that $S \cap (T \cup U) = (S \cap T) \cup (S \cap U)$.

Solution. asdf

(b) Prove that $S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$.

Solution. asdf

Taken together, these results are called the **distributive properties** of the set operations.

3. **Power sets.** Recall that the power set $\mathcal{P}(S)$ of a set S is the set whose elements are all the subsets of S .

- (a) Determine whether the following statement is true or false:

The power set of a set contains at least two distinct elements.

If the statement is true, prove it. If the statement is false, disprove it by providing a counterexample.

Hint. Revisit the proofs of [Lemma 1.1.11 Power sets are non-empty](#).

Solution. asdf

Text after the parts.

4. **Set operations and containment.** Let S and T be sets.

- (a) Prove that $S \cap T = S$ if and only if $S \subseteq T$.

Solution. Since $S \cap T \subseteq S$, it suffices to show that $S \subseteq S \cap T$ if and only if $S \subseteq T$. To that end, note that if $S \subseteq S \cap T$, then (1) of [Proposition 1.1.14 Set operations and containment](#) implies that $S \subseteq T$.

Conversely, we note that $S \subseteq S$, so that if $S \subseteq T$, then (1) of [Proposition 1.1.14 Set operations and containment](#) implies that $S \subseteq$

$$S \cap T.$$

- (b) Prove that $S \cup T = S$ if and only if $T \subseteq S$.

Solution. Since $S \cup S \cup T$, it suffices to show that $S \cup T \subseteq S$ if and only if $T \subseteq S$. To that end, note that if $S \cup T \subseteq S$, then (2) of [Proposition 1.1.14 Set operations and containment](#) implies that $T \subseteq S$.

Conversely, we note that $S \subseteq S$, so that if $T \subseteq S$, then (2) of [Proposition 1.1.14 Set operations and containment](#) implies that $S \cup T \subseteq S$.

- (c) Prove that $S \setminus T = S$ if and only if $S \cap T = \emptyset$.

Solution. First suppose that $S \cap T = \emptyset$. Then $x \notin T$ for all $x \in S$, and so

$$S \setminus T = \{x \mid x \in S \text{ and } x \notin T\} = \{x \mid x \in S\} = S.$$

Conversely, now suppose that $S \cap T \neq \emptyset$. Then $x \in S \cap T$, and so $x \in S$ and $x \in T$. Since $x \in T$, $x \notin S \setminus T$. So x is an element of S but not $S \setminus T$; we conclude that $S \setminus T \neq S$.

In-depth solutions to these reading questions are available. However, these are provided in order to check your work. If you are unable to solve the above reading questions, discuss the problem with classmates and/or instructor rather than spoiling the answers.

1.2 Maps and Functions

In this section, we introduce the fundamental notion of a map between sets. Informally, a map is an assignment of an element of one set to each element of another. You probably have been exposed to maps in previous math courses in the form of real-valued functions of one real variable. We will treat maps in far greater generality, which will allow us to perform mathematical analysis in a wider array of spaces.

1.2.1 Introduction to Maps

First, however, we define the **Cartesian product** of two sets. This relies on a notion of an ordered pair of objects x and y , which we denote (x, y) . Like formalized set theory, rigorous constructions of ordered pairs are, in this context, unnecessarily complicated and annoying to work with. As such, we will take them to be well-defined. The salient properties of an ordered pair (x, y) are its **entries** x and y , which are called its first and second entries, respectively.

Definition 1.2.1 Cartesian product. The **Cartesian product** $X \times Y$ of two sets X and Y is the set which contains all ordered pairs whose first entry is an element of X and whose second entry is an element of Y ; that is,

$$X \times Y = \{(x, y) \mid x \in X \text{ and } y \in Y\}.$$

◆

Example 1.2.2 An example of Cartesian products. Let $X = \{1, 2\}$ and $Y = \{3, 4, 5\}$. Then

$$X \times Y = \{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)\}.$$

Note that $X \times Y \neq X \cup Y$. ▲

Recall that we would like a map between two sets to be an association of the elements of one set to the elements of the other. We will model this association via ordered pairs. To that end, a map between two sets will just be a subset of their Cartesian product which satisfies certain existence and uniqueness properties.

Definition 1.2.3 Map; function. A subset $f \subseteq X \times Y$ of the Cartesian product $X \times Y$ of two sets X and Y is called a **map** from X (which is called the **domain** of f and whose elements are called **inputs**) to Y (which is called the **codomain** of f and whose elements are called **outputs**) if it satisfies the following conditions:

1. **Totality:** For all inputs $x \in X$, there is an output $y \in Y$ so that $(x, y) \in f$.
2. **Uniqueness:** For all inputs $x \in X$ and outputs $y_1, y_2 \in Y$, if $(x, y_1), (x, y_2) \in f$, then $y_1 = y_2$.

So f is a map from X to Y if and only if for each input $x \in X$, there is a unique output $y \in Y$ so that $(x, y) \in f$. We will write $f: X \rightarrow Y$ to indicate that f is a map from X to Y , and we will write $f: x \mapsto y$ to indicate that $(x, y) \in f$. The set of maps from X to Y is denoted $Y^X \subseteq X \times Y$.

We will call a map $f: X \rightarrow Y$ whose codomain Y is a number system like the natural numbers $Y = \mathbb{N}$, the integers $Y = \mathbb{Z}$, the rational numbers $Y = \mathbb{Q}$, the real numbers $Y = \mathbb{R}$, or the complex numbers $Y = \mathbb{C}$ a Y -valued **function** on X . We will investigate the natural numbers $\mathbb{N}$ in [provisional cross-reference: section-peano-arithmetic-and-the-natural-numbers], and [provisional cross-reference: chapter-constructing-the-number-systems] will be dedicated to constructing the other number systems with which we will become acquainted throughout the course of this text. ◆

So a map between sets is just a subset of their Cartesian product such that each element of the domain appears in precisely one pair with an element of the codomain. The above existence and uniqueness properties of a map implies that there is an unambiguous association between the elements of the domain and the elements of the codomain.

Example 1.2.4 Examples of maps. The following are examples of maps between sets:

1. Let $X = \{1, 2\}$ and $Y = \{\text{red}, \text{blue}\}$, and consider the following subsets of $X \times Y$:

$$\begin{aligned} f_1 &= \{(1, \text{red}), (2, \text{red})\} \\ f_2 &= \{(1, \text{red}), (2, \text{blue})\} \\ f_3 &= \{(1, \text{blue}), (2, \text{red})\} \\ f_4 &= \{(1, \text{blue}), (2, \text{blue})\}. \end{aligned}$$

As defined above, f_1, f_2, f_3 , and f_4 are maps from X to Y . In fact, these are the only such maps; that is,

$$Y^X = \{f_1, f_2, f_3, f_4\}.$$

2. The empty set $\emptyset: \emptyset \rightarrow Y$ is the unique map from the empty set $\emptyset$ to any set Y ; that is,

$$Y^\emptyset = \{\emptyset\}.$$

Note that none of the elements of the codomain Y are associated with elements of the empty domain $\emptyset$. However, this does not imply that $\emptyset$ is not a map; in general, not all elements of the codomain of a map need be associated with elements of the domain, since *the existence property applies only to the domain*.

3. Let $U \subseteq X$ be a subset of a set X . The **inclusion map** $\iota_U^X: U \rightarrow X$ of U in X is the map from U to X defined by

$$\iota_U^X = \{(u, x) \in U \times X \mid u = x\}.$$

That is, $\iota_U^X(u) = u$ for all $u \in U$.

The inclusion map ι_X^X of X in itself is called the **identity map** on X , and is denoted $\text{id}_X: X \rightarrow X$.



Remark 1.2.5 Passive and active characterizations of maps. It may be difficult to see how the above definition of a map is a generalization of the notion of a real-valued function of a real variable. In fact, the above definition might seem more closely related to the graph of such a function, since the uniqueness property of maps is just a generalization of the vertical line test for functions. We will refer to this perspective on maps as subsets of a Cartesian product satisfying certain abstract properties as the **passive characterization** of what maps *are*. In contrast, we often like to think of a map as an active process, which takes an input and transforms it into the unique output with which it is associated. We will refer to this perspective as the **active characterization** of what maps *do*. When it benefits us to favor one perspective over another, we will emphasize either the active or passive characterization over the other. The following definitions allow us to convert between these two equivalent characterizations of maps.

Definition 1.2.6 Image; pre-image. Let $f: X \rightarrow Y$ be a map from a set X to a set Y . The **image** $f(x)$ of an input $x \in X$ under f is the unique output so that $(x, f(x)) \in Y$. The **image** $f(U)$ of a subset $U \subseteq X$ is the subset of the codomain Y defined by

$$f(U) = \{f(x) \in Y \mid x \in U\}.$$

The image $f(X)$ of X itself is called the **image** of f , and is denoted $\text{im}(f) = f(X)$.

The **pre-image** $f^{-1}(V)$ of a subset $V \subseteq Y$ of the codomain Y under f is the set of all inputs whose image lies in V ; that is,

$$f^{-1}(V) = \{x \in X \mid f(x) \in V\}.$$

The **pre-image** $f^{-1}(y)$ of an output $y \in Y$ under f is just the pre-image of the singleton set $\{y\}$ under f ; that is,

$$f^{-1}(y) = f^{-1}(\{y\}) = \{x \in X \mid f(x) = y\}$$



Any map between sets induces a map from any subset of its domain to its codomain via a process known as **restriction**. In light of the active charac-

terization of what maps do, this may seem almost trivial. However, we will formalize this notion of restriction in terms of the passive characterization of what maps are in order to prove rigorously our intuitions are correct. You may be better served by conceptualizing the restriction of a map to be the same active process of transforming inputs into outputs but with a smaller domain, rather than the passive definition below.

Definition 1.2.7 Restriction. The **restriction** $f|_U$ of a map $f: X \rightarrow Y$ between sets X and Y to a subset $U \subseteq X$ of its domain X is the subset of the Cartesian product $U \times Y$ defined by

$$\begin{aligned} f|_U &= \{(u, y) \in U \times Y \mid f(u) = y\} = \{(u, y) \in U \times Y \mid (u, y) \in f\} \\ &= (U \times Y) \cap f. \end{aligned}$$

◆

As our intuition about the active characterization of what maps do tells us, the restriction of a map to a subset of its domain is itself a map from the chosen subset to the domain of the original map.

Proposition 1.2.8 Restrictions are maps. *Let $f: X \rightarrow Y$ be a map from a set X to a set Y . For any subset $U \subseteq X$, the restriction $f|_U$ of f to U is a map from U to Y .*

Proof. Let $u \in U$. Then $u \in X$ is an input to f , and so $(u, y) \in f$ for some output $y \in Y$. We conclude that

$$(x, y) \in \{(u, y) \in U \times Y \mid (u, y) \in f\} = f|_U,$$

so that $f|_U$ satisfies the totality condition.

Let $u \in U$ and $y_1, y_2 \in Y$, and suppose that $(u, y_1), (u, y_2) \in f|_U$. Since

$$f|_U = (U \times Y) \cap f \subseteq f,$$

we have that $(u, y_1), (u, y_2) \in f$, and so $y_1 = y_2$; that is, $f|_U$ satisfies the uniqueness condition. Since $f|_U$ satisfies the totality condition and the uniqueness condition, it is a map from U to Y . ■

We will shortly see another way to construct restrictions as compositions with inclusion maps.

1.2.2 Composition and Invertibility

Viewed through the perspective of the active characterization of what maps do, it may seem obvious that we can combine two maps between sets by first acting on it with one map and then acting on its image by the other map. This new process yields the **composition** of the two maps, and it may be intuitively clear that this composition is itself a map. However, we will formalize this notion of composition in the passive characterization, so that we may prove this fact rigorously. Again, it may better serve you to conceptualize the composition of maps in the context of the active characterization of what maps do rather than the passive characterization of what maps are in order to better inform your intuition about composition.

Definition 1.2.9 Composition. Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be maps between sets X , Y , and Z . The **composition** $g \circ f$ of g with f is the map from X to Z defined by the formula

$$(g \circ f)(x) = g(f(x)).$$

We aren't restricted to the composition of just two maps. In fact, we can compose any finite number of maps with compatible domains and codomains by iterating the above construction. ◆

Proposition 1.2.10 Associativity of composition. *Let $f: W \rightarrow X$, $g: X \rightarrow Y$, and $h: Y \rightarrow Z$ be maps between sets W , X , Y , and Z . Then*

$$(h \circ g) \circ f = h \circ (g \circ f).$$

Proof. Note that

$$\begin{aligned} ((h \circ g) \circ f)(w) &= (h \circ g)(f(w)) = h(g(f(w))) \\ &= h((g \circ f)(w)) = (h \circ (g \circ f))(w) \end{aligned}$$

for all $w \in W$. Since $(h \circ g) \circ f$ and $h \circ (g \circ f)$ agree on all inputs $w \in W$,

$$(h \circ g) \circ f = h \circ (g \circ f).$$

The above notion of composition introduces a natural question about maps between sets: ■

This question of map **invertibility** is fundamental, and will recur throughout this text.

Definition 1.2.11 Invertibility. Let $f: X \rightarrow Y$ be a map from a set X to a set Y . A map $g: Y \rightarrow X$ is a **left inverse** to f if $g \circ f = \text{id}_X$, and g is called a **right inverse** to f if $f \circ g = \text{id}_Y$. g is called an **inverse** to f if it is both a left and right inverse to f ; that is, g is an inverse to f if $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$.

A map is called **left-invertible** (resp. **right-invertible**, **invertible**) if it has a left inverse (resp. right inverse, inverse). ◆

Example 1.2.12 Examples of inverses. The following are examples of left inverses, right inverses, and inverses of maps.

1. Let $\emptyset \subsetneq U \subseteq X$ be a non-empty subset of a set X , and fix an element $u \in U$. Consider the constant map $c: X \rightarrow U$ from X to U defined piecewise by the formula

$$c(x) = \begin{cases} x, & x \in U \\ u, & x \notin U. \end{cases}$$

Note that

$$(c \circ \iota_U^X)(u) = c(\iota_U^X(u)) = c(u) = u = \text{id}_U(u)$$

for all $u \in U$, and so $c \circ \iota_U^X = \text{id}_U$; that is, c is a left inverse to the inclusion map $\iota_U^X: U \rightarrow X$ of U in X .

2. Fix sets X and Y , and suppose that $Y \neq \emptyset$ is non-empty. Let $\pi: X \times Y \rightarrow X$ be the **projection map** defined by the formula $\pi(x, y) = x$. Fix an element $y \in Y$, and consider the map $c: X \rightarrow X \times Y$ from X to $X \times Y$ defined by the formula $c(x) = (x, y)$. Then

$$(\pi \circ c)(x) = \pi(c(x)) = \pi(x, y) = x = \text{id}_X(x)$$

for all $x \in X$, and so $\pi \circ c = \text{id}_X$; that is, c is a right inverse to the projection map π .

3. The identity map $\text{id}_X: X \rightarrow X$ on a set X is its own inverse, since $\text{id}_X \circ \text{id}_X = \text{id}_X$. This is actually a special case of the first example above, where we take $U = X$. More generally, $f \circ \text{id}_X = f$ for all maps f with domain X , and $\text{id}_X \circ g = g$ for all maps g with codomain X .



We now give alternative characterizations of left-invertible, right-invertible, and invertible maps. Intuitively, a map is left-invertible if it does not associate two distinct inputs with the same output, and a map is right-invertible if each output is associated with at least one input. These properties mirror the totality and uniqueness conditions for maps, and will turn out to together imply the existence of a two-sided inverse, which is both a left and right inverse.

Definition 1.2.13 Injectivity; surjectivity; bijectivity. Let $f: X \rightarrow Y$ be a map from a set X to a set Y .

1. f is called **injective** (or **one-to-one**) if for all inputs $x_1, x_2 \in X$, if $f(x_1) = f(x_2)$, then $x_1 = x_2$. In this case, f is also called an **injection** of X into Y .
2. f is called **surjective** (or **onto**) if for all outputs $y \in Y$ there is an input $x \in X$ so that $y = f(x)$. In this case, f is also called a **surjection** of X onto Y .
3. f is called **bijective** if it is both injective and surjective. In this case, f is also called a **bijection** or a **one-to-one correspondence** between X and Y .



Example 1.2.14 Examples of injections, surjections, and bijections. The following are examples of injective, surjective, and bijective functions.

1. Let $U \subseteq X$ be a subset of a set X , and consider $u_1, u_2 \in U$. If $\iota_U^X(u_1) = \iota_U^X(u_2)$, then

$$u_1 = \iota_U^X(u_1) = \iota_U^X(u_2) = u_2.$$

So the inclusion map $\iota_U^X: U \rightarrow X$ is injective.

2. Fix sets X and Y , and suppose that $Y \neq \emptyset$ is non-empty. Let $\pi: X \times Y \rightarrow X$ be the **projection map** defined by the formula $\pi(x, y) = x$.

Let $x \in X$, and note that since $Y \neq \emptyset$ is non-empty, there is some element $y \in Y$. We note that $\pi(x, y) = x$. Since each output $x \in X$ is the image of some input under π , π is surjective.

3. The identity map $\text{id}_X: X \rightarrow X$ on a set X is injective by virtue of being an inclusion map. id_X is also surjective, since $\text{id}_X(x) = x$ for all $x \in X$. Thus id_X is bijective.



It is no accident that the examples for left- and right-invertibility coincided with the examples for injectivity and surjectivity. It turns out that injectivity is essentially equivalent to left-invertibility, and surjectivity is equivalent to right-invertibility.

Theorem 1.2.15 Invertibility. Let $f: X \rightarrow Y$ be a map from a set X to a set Y .

1. If f is left-invertible, then it is injective.

2. If $X \neq \emptyset$ is non-empty and f is injective, then it is left-invertible.
3. If f is right-invertible, then it is surjective.

Proof.

1. Suppose that f is left-invertible, so that $g \circ f = \text{id}_X$ for some map $g: Y \rightarrow X$. That f is injective follows from the injectivity of the identity map id_X and (d) of [provisional cross-reference: problem-compositions-of-injections-surjections-and-bijections].
2. Conversely, now suppose that $X \neq \emptyset$ is non-empty and f is injective. Then for each output $y \in Y$, there is at most one input $x \in X$ so that $y = f(x)$. Specifically, for each $y \in \text{im}(f)$, there is a unique input $x \in X$ so that $y = f(x)$. Denote by $g(y) = x$ this unique input.

Since $X \neq \emptyset$ is non-empty, it contains an element $x \in X$. Define $g(y) = x$ for all $y \in Y \setminus \text{im}(f)$. We have defined a map $g: Y \rightarrow X$. Note that

$$(g \circ f)(x) = g(f(x)) = x = \text{id}_X(x)$$

for all inputs $x \in X$, and so $g \circ f = \text{id}_X$; that is, g is a left inverse to f , and so f is left-invertible.

3. Finally, suppose that f is right-invertible, so that $f \circ g = \text{id}_Y$ for some map $g: Y \rightarrow X$. That f is surjective follows from the surjectivity of the identity map id_Y and (e) of [provisional cross-reference: problem-compositions-of-injections-surjections-and-bijections].

■

Proving the remaining direction of the equivalence between right invertibility and surjectivity requires a strong assumption about set theory called the **axiom of choice**, which we do not yet have the language to express in its simplest form. We will talk about the axiom of choice shortly, when we generalize the Cartesian product of two sets using indexing maps.

Corollary 1.2.16 Invertibility. *A map is invertible if and only if it is bijective.*

Proof. Let $f: X \rightarrow Y$ be a map from a set X to a set Y . If f is invertible, it is both left- and right-invertible, and so f is both injective and surjective by (1) and (3) of [Theorem 1.2.15 Invertibility](#). Hence f is bijective.

Conversely, now suppose that f is bijective, so that f is both injective and surjective. If $X = \emptyset$, then

$$\begin{aligned} Y &= \text{im}(f) = f(X) = f(\emptyset) \\ &= \{f(x) \in Y \mid x \in \emptyset\} = \emptyset \end{aligned}$$

by the surjectivity of f . Thus $f: \emptyset \rightarrow \emptyset$ is the unique map from $\emptyset$ to itself, the identity map $\text{id}_\emptyset: \emptyset \rightarrow \emptyset$. In particular, f is invertible.

So we may assume that $X \neq \emptyset$ is non-empty. Since f is injective, it is left-invertible by (2) of [Theorem 1.2.15 Invertibility](#). So $g \circ f = \text{id}_X$ for some map $g: Y \rightarrow X$; that is, $g(f(x)) = x$ for all inputs $x \in X$. It suffices to show that g is also a right inverse to f .

To that end, consider an output $y \in Y$. Since f is surjective, $y = f(x)$ for some input $x \in X$. Then

$$(f \circ g)(y) = f(g(x)) = f(g(f(x))) = f(x) = y = \text{id}_Y(y).$$

Hence $f \circ g = \text{id}_Y$, and so g is an inverse to f ; that is, f is invertible. ■

We will investigate the invertibility and bijectivity of maps and the consequences these properties have on the domains and codomains of such maps more at the end of this chapter, when we study **cardinality** and the size of sets. For now, however, we will move on to an important convention which is ubiquitous throughout mathematics: the notions of indexed sets and families.

1.2.3 Indexing

Considered as an association between sets, a map can be thought to index its image by the elements of its domain. That is, a map can transfer information from its domain to its image. Just what this transferred information looks like can vary tremendously, but suffice it to say that this notion of indexing is fundamental and will recur throughout this text.

Definition 1.2.17 Indexed family; indexed set. A set X is **indexed** by a set $\mathcal{J}$ if there is a surjection $x: \mathcal{J} \rightarrow X$. $\mathcal{J}$ is called an **indexing set**, and for each input $j \in \mathcal{J}$, called an **index**, its image is denoted $x_j = x(j)$.

The map x is called an **(indexed) family** of elements of X , and is denoted $(x_j)_{j \in \mathcal{J}}$.

Consider a map $x: \mathcal{J} \rightarrow X$ of a set $\mathcal{J}$, called the **indexing set** and whose elements are called **indices**, onto a set X . The image of such an index $j \in \mathcal{J}$ under such an **(indexed) family** x is often denoted $x_j = x(j)$, and x is often denoted $(x_j)_{j \in \mathcal{J}}$. We say that $\text{im}(x)$ is **indexed** by $\mathcal{J}$, and we write

$$\{x_j\}_{j \in \mathcal{J}} = \{x_j \in X \mid j \in \mathcal{J}\} = \text{im}(x).$$

◆

We can also use this notion of indexing to generalize the intersection and union of sets to act on arbitrary collections of sets. This generalization of the intersection and union of two sets relies on the natural similarities between logical conjunction $\wedge$ and universal quantification $\forall$ and between logical disjunction $\vee$ and existential quantification $\exists$.

Definition 1.2.18 Intersection; union. Let $(U_j)_{j \in \mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X indexed by a set $\mathcal{J}$. The **intersection** $\bigcap_{j \in \mathcal{J}} U_j$ of this indexed family is the subset of X defined by

$$\bigcap_{j \in \mathcal{J}} U_j = \{x \in X \mid x \in U_j \text{ for all } j \in \mathcal{J}\}.$$

The **union** $\bigcup_{j \in \mathcal{J}} U_j$ of this indexed family is the subset of X defined by

$$\bigcup_{j \in \mathcal{J}} U_j = \{x \in X \mid x \in U_j \text{ for some } j \in \mathcal{J}\}.$$

◆

These generalizations of the intersection and union of two sets extend naturally to generalizations of each of the results proven in the previous section.

Theorem 1.2.19 Analogous properties of indexed intersections and unions. Let $(U_j)_{j \in \mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X indexed by a set $\mathcal{J}$.

1. Analogously to [Proposition 1.1.14 Set operations and containment](#), for any subset $V \subseteq X$, $V \subseteq \bigcap_{j \in \mathcal{J}} U_j$ if and only if $V \subseteq U_j$ for all indices

$j \in \mathcal{J}$, and $\bigcup_{j \in \mathcal{J}} U_j \subseteq V$ if and only if $U_j \subseteq V$ for all indices $j \in \mathcal{J}$.

2. Analogously to (2) of [Corollary 1.1.15 Properties of the set operations](#),

$$\bigcap_{k \in \mathcal{K}} U_{f(k)} = \bigcap_{j \in \mathcal{J}} U_j$$

and

$$\bigcup_{k \in \mathcal{K}} U_{f(k)} = \bigcup_{j \in \mathcal{J}} U_j$$

for any set $\mathcal{K}$ and bijection $f: \mathcal{K} \rightarrow \mathcal{J}$.

3. Analogously to [\[provisional cross-reference: problem-distributivity-of-set-operations\]](#),

$$V \cup \bigcap_{j \in \mathcal{J}} U_j = \bigcap_{j \in \mathcal{J}} (V \cup U_j)$$

and

$$V \cap \bigcup_{j \in \mathcal{J}} U_j = \bigcup_{j \in \mathcal{J}} (V \cap U_j)$$

for any subset $V \subseteq X$.

4. Analogously to [\[provisional cross-reference: problem-de-morgans-laws\]](#),

$$V \setminus \bigcap_{j \in \mathcal{J}} U_j = \bigcup_{j \in \mathcal{J}} (V \setminus U_j)$$

and

$$V \setminus \bigcup_{j \in \mathcal{J}} U_j = \bigcap_{j \in \mathcal{J}} (V \setminus U_j)$$

for any subset $V \subseteq X$.

You will have an opportunity to prove the above results in the problems for this section.

Text after the last subsection.

1.2.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

1. **Images and pre-images.** Let $f_4: \{1, 2\} \rightarrow \{\text{red}, \text{blue}\}$ be the map from $\{1, 2\}$ to $\{\text{red}, \text{blue}\}$ defined in (2) of [\[provisional cross-reference: example-maps\]](#); that is,

$$f_4 = \{(1, \text{blue}), (2, \text{blue})\}.$$

Compute the given images and pre-images. Express your answer in roster notation.

- (a) Compute the images $f_4(1)$ and $f_4(2)$.

Answer. $f_4(1) = f_4(2) = \text{blue}$.

Solution. Since $(1, \text{blue}) \in f_4$, the image $f_4(1)$ of 1 under f_4 is

$$f_4(1) = \text{blue}.$$

Similarly, since $(2, \text{blue}) \in f_4$, the image $f_4(2)$ of 2 under f_4 is

$$f_4(2) = \text{blue}.$$

- (b) Compute the image $\text{im}(f_4)$.

Answer. $\text{im}(f_4) = \{\text{blue}\}$.

Solution. Since $f_4(x) = \text{blue}$ for all $x \in \{1, 2\}$, the image $\text{im}(f_4)$ of f_4 is

$$\text{im}(f_4) = f_4(\{1, 2\}) = \{\text{blue}\}.$$

- (c) Compute the pre-images $f_4^{-1}(\text{red})$ and $f_4^{-1}(\text{blue})$.

Answer. $f_4^{-1}(\text{red}) = \emptyset$ and $f_4^{-1}(\text{blue}) = \{1, 2\}$.

Solution. Since $f_4(x) = \text{blue}$ for all $x \in \{1, 2\}$, the pre-images $f_4^{-1}(\text{red})$ and $f_4^{-1}(\text{blue})$ of red and blue under f_4 are

$$f_4^{-1}(\text{red}) = \{x \in \{1, 2\} \mid f_4(x) = \text{red}\} = \emptyset$$

and

$$f_4^{-1}(\text{blue}) = \{x \in \{1, 2\} \mid f_4(x) = \text{blue}\} = \{1, 2\}.$$

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.2.5 Reading Questions

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Composition and “jectivity”.** Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be maps between sets X , Y , and Z .

- (a) Prove that if f and g are injective, then so is their composition $g \circ f: X \rightarrow Z$.

Solution. Let $x_1, x_2 \in X$, and suppose that $(g \circ f)(x_1) = (g \circ f)(x_2)$. Then

$$g(f(x_1)) = (g \circ f)(x_1) = (g \circ f)(x_2) = g(f(x_2)),$$

and so the injectivity of g implies that $f(x_1) = f(x_2)$. Moreover, the injectivity of f implies that $x_1 = x_2$. In summary, we have

shown that for all inputs $x_1, x_2 \in X$, if $(g \circ f)(x_1) = (g \circ f)(x_2)$, then $x_1 = x_2$; that is, $g \circ f$ is injective.

- (b) Prove that if f and g are surjective, then so is their composition $g \circ f: X \rightarrow Z$.

Solution. Let $z \in Z$. Since g is surjective, $z = g(y)$ for some $y \in Y$. Moreover, since f is surjective, $y = f(x)$ for some $x \in X$. Thus

$$z = g(y) = g(f(x)) = (g \circ f)(x).$$

In summary, we have shown that for every output $z \in Z$, there is some input $x \in X$ so that $z = (g \circ f)(x)$; that is, $g \circ f$ is surjective.

- (c) Prove that if f and g are bijective, then so is their composition $g \circ f: X \rightarrow Z$.

Solution. Since f and g are bijective, they are both injective and surjective. (1) and (2) above not imply that their composition $g \circ f$ is both injective and surjective; that is, $g \circ f$ is bijective.

- (d) Prove that if the composition $g \circ f: X \rightarrow Z$ is injective, then so is f .

Hint. Prove the desired result by contraposition.

Solution. Suppose that f is not injective. Then $f(x_1) = f(x_2)$ for some distinct inputs $x_1, x_2 \in X$. Note that

$$(g \circ f)(x_1) = g(f(x_1)) = g(f(x_2)) = (g \circ f)(x_2).$$

Since $(g \circ f)(x_1) = (g \circ f)(x_2)$ but $x_1 \neq x_2$, $g \circ f$ is not injective.

In summary, we have shown that if f is not injective, then $g \circ f$ is not injective. The contrapositive of this implication is the desired result: if $g \circ f$ is injective, then so is f .

- (e) Prove that if the composition $g \circ f: X \rightarrow Z$ is surjective, then so is g .

Hint. Prove the desired result by contraposition.

Solution. Suppose that g is not surjective. Then there is some $z \in Z$ so that $z \neq g(y)$ for all $y \in Y$. Note that

$$(g \circ f)(x) = g(f(x)) \neq z$$

for all $x \in X$, and so $g \circ f$ is not surjective.

In summary, we have shown that if g is not surjective, then $g \circ f$ is not surjective. The contrapositive of this implication is the desired result: if $g \circ f$ is surjective, then so is g .

In-depth solutions to these reading questions are available. However, these are provided in order to check your work. If you are unable to solve the above reading questions, discuss the problem with classmates and/or instructor rather than spoiling the answers.

1.3 Peano Arithmetic and the Natural Numbers

In this section, we formally introduce the natural number system $\mathbb{N}$. Also called the counting numbers, the natural numbers form the simplest of the number systems; in fact, many people would date the advent of mathematics itself to the invention/discovery of these numbers.

1.3.1 Axiomatizing the Natural Numbers

There are many nonequivalent formalizations and models of the natural number system, but the standard axiomatization is due to Giuseppe Peano in the 19th century. Peano arithmetic is nonconstructive; it gives a description of the natural number system but not a model. We won't introduce any specific model of the natural number system, but, rest assured, there are several. For our purposes, $\mathbb{N}$ itself can be any set which conforms to the following characterization:

Definition 1.3.1 Natural numbers. The **natural number system** $\mathbb{N}$ is a nonempty set with an element $0 \in \mathbb{N}$, called **zero**, along with functions $\text{succ}: \mathbb{N} \rightarrow \mathbb{N}$ and $\text{add}, \text{mult}: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, called the **successor function** and the **addition** and **multiplication operations** and a subset $\text{comp} \subseteq \mathbb{N} \times \mathbb{N}$ of the Cartesian product $\mathbb{N} \times \mathbb{N}$ of $\mathbb{N}$ with itself, called the **standard ordering**, which satisfy the following properties:

1. The successor $\text{succ}(n)$ of any natural number $n \in \mathbb{N}$ is non-zero.
2. The successor function $\text{succ}: \mathbb{N} \rightarrow \mathbb{N}$ is injective.
3. $\text{add}(n, 0) = n$ for all natural numbers $n \in \mathbb{N}$.
4. $\text{add}(m, \text{succ}(n)) = \text{succ}(\text{add}(m, n))$ for all natural numbers $m, n \in \mathbb{N}$.
5. $\text{mult}(n, 0) = 0$ for all natural numbers $n \in \mathbb{N}$.
6. $\text{mult}(m, \text{succ}(n)) = \text{add}(\text{mult}(m, n), m)$ for all natural numbers $n \in \mathbb{N}$.
7. For all natural numbers $l, m \in \mathbb{N}$, $(l, m) \in \text{comp}$ if and only if $\text{add}(l, n) = m$ for some natural number $n \in \mathbb{N}$.
8. For all subsets $U \subseteq \mathbb{N}$, $U = \mathbb{N}$ if and only if it satisfies the following conditions:
 - (a) $0 \in U$.
 - (b) For all natural numbers $n \in \mathbb{N}$, if $n \in U$, then $\text{succ}(n) \in U$.

Typically, for all natural numbers $m, n \in \mathbb{N}$ we write $m + n = \text{add}(m, n)$ and $mn = m \cdot n = \text{mult}(m, n)$, and we write $m \leq n$ if and only if $(m, n) \in \text{comp}$. The successor of 0 is denoted $1 = \text{succ}(0)$. Likewise, the successor of 1 is denoted $2 = \text{succ}(1)$, the successor of 2 is denoted $3 = \text{succ}(2)$, and so on. We see that

$$\text{succ}(n) = \text{succ}(\text{add}(n, 0)) = \text{add}(n, \text{succ}(0)) = n + 1$$

for all natural numbers $n \in \mathbb{N}$. With these new notations, we may restate the Peano axioms as follows:

1. $n + 1 \neq 0$ for all $n \in \mathbb{N}$.
2. For all natural numbers $m, n \in \mathbb{N}$, if $m + 1 = n + 1$, then $m = n$.

3. $n + 0 = n$ for all natural numbers $n \in \mathbb{N}$.
4. $m + (n + 1) = (m + n) + 1$ for all natural numbers $m, n \in \mathbb{N}$.
5. $n \cdot 0 = 0$ for all natural numbers $n \in \mathbb{N}$.
6. $m(n + 1) = mn + m$ for all natural numbers $n \in \mathbb{N}$.
7. For all natural numbers $l, m \in \mathbb{N}$, $l \leq m$ if and only if $l + n = m$ for some natural number $n \in \mathbb{N}$.
8. For all subsets $U \subseteq \mathbb{N}$, $U = \mathbb{N}$ if and only if it satisfies the following conditions:
 - (a) $0 \in U$.
 - (b) For all natural numbers $n \in \mathbb{N}$, if $n \in U$, then $n + 1 \in U$.

◆

Under the usual (and original) interpretation of the natural number system $\mathbb{N}$ as characterizing quantities of discrete physical objects, the above rules (with the possible exception of the final axiom, called the **axiom of induction**) may seem obviously true but also somewhat arbitrary. It is a fact, though we will not prove it here, that these axioms are independent of each other; it is impossible to derive one of the above statements from only the others. We hope that this, along with the results in this section about arithmetic, convince you that the above axiomatization is a good choice, in that it is efficient in the above sense and that it can be used to prove many arithmetical truths.

First, however, we introduce some ubiquitous terminology and notation. A family $(x_n)_{n \in \mathcal{N}}$ of objects x_n indexed by some subset $\mathcal{N} \subseteq \mathbb{N}$ is called a **sequence**. If the index set $\mathcal{N}$ is of the form

$$\mathcal{N} = \{n \in \mathbb{N} \mid a \leq n \text{ and } n \leq b\}$$

for some natural numbers $a, b \in \mathbb{N}$, then we denote this family by $(x_n)_{n=a}^b$. Similarly, if the index set $\mathcal{N}$ is of the form

$$\mathcal{N} = \{n \in \mathbb{N} \mid a \leq n\}$$

for some natural number $a \in \mathbb{N}$, then we denote this family by $(x_n)_{n=a}^\infty$. These notations also apply to intersections and unions of sequences of sets indexed by subsets of $\mathbb{N}$ of the above form.

Lemma 1.3.2 0 is minimal. *For all natural numbers $n \in \mathbb{N}$, if $n \leq 0$, then $n = 0$.*

Proof. We first observe that $\{0\} \cup \text{im}(\text{succ}) = \mathbb{N}$. Indeed, note that $0 \in \{0\} \subseteq \{0\} \cup \text{im}(\text{succ})$. Moreover, if $n \in \{0\} \cup \text{im}(\text{succ})$, then

$$\text{succ}(n) \in \text{im}(\text{succ}) \subseteq \{0\} \cup \text{im}(\text{succ}).$$

Thus $\{0\} \cup \text{im}(\text{succ}) = \mathbb{N}$ by the axiom of induction.

Let $n \in \mathbb{N}$ be a natural number, and suppose that $n \leq 0$ for some natural number $n \in \mathbb{N}$. Then $n + m = 0$ for some natural number $m \in \mathbb{N}$. Suppose for a contradiction that $m = p + 1$ for some natural number $p \in \mathbb{N}$. Then

$$0 = n + m = n + (p + 1) = (n + p) + 1,$$

which contradicts the first Peano axiom. Thus $m \notin \text{im}(\text{succ})$. However, $m \in \mathbb{N} = \{0\} \cup \text{im}(\text{succ})$, and so $m \in \{0\}$; that is, $m = 0$, and so

$$n = n + 0 = n + m = 0.$$

■

Proposition 1.3.3 Principle of strong induction. *For all subsets $U \subseteq \mathbb{N}$, $U = \mathbb{N}$ if and only if it satisfies the following conditions:*

1. $0 \in U$.
2. For all natural numbers $n \in \mathbb{N}$, if $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n$, then $n + 1 \in U$.

Proof. Clearly $\mathbb{N}$ satisfies the above conditions, and so it suffices to show that any subsets which satisfies these conditions is equal to $\mathbb{N}$. To that end, let $U \subseteq \mathbb{N}$ be a subset which satisfies the above conditions, and consider the subset

$$V = \{n \in \mathbb{N} \mid k \in U \text{ for all } k \in \mathbb{N} \text{ so that } k \leq n\}.$$

Since the only $k \in \mathbb{N}$ so that $k \leq 0$ is $k = 0$ itself, and $0 \in U$, we have that $0 \in V$. Moreover if $n \in V$, then $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n$, and so $n + 1 \in U$ by hypothesis. It remains to conclude that $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n + 1$.

To that end, let $k \in \mathbb{N}$, and suppose that $k \leq n + 1$. Then $k + m = n + 1$ for some natural number $m \in \mathbb{N}$. If $m = 0$, then $k = n + 1 \in U$. On the other hand, if $m \neq 0$, then the argument in the proof of [Lemma 1.3.2 0 is minimal](#) implies that $m = p + 1$ for some natural number $p \in \mathbb{N}$. Thus

$$(k + p) + 1 = k + (p + 1) = k + m = n + 1,$$

and so the injectivity of the successor function implies that $k + p = n$; that is, $k \leq n$ and so $k \in U$. Since $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n + 1$, we conclude that $n + 1 \in V$.

In summary, we have shown that $0 \in V$ and that for all natural numbers $n \in \mathbb{N}$, if $n \in V$, then $n + 1 \in V$. Thus $V = \mathbb{N}$ by the axiom of induction. One can check that $\mathbb{N} = V \subseteq U \subseteq \mathbb{N}$, and so $U = \mathbb{N}$. ■

The axiom of induction and [Proposition 1.3.3 Principle of strong induction](#) form the basis for ubiquitous methods of proof called **proof by induction** and **proof by strong induction**, respectively. You will have the opportunity to practice constructing such proofs in the problems for this section.

1.3.2 Properties of Arithmetic

The axioms of Peano arithmetic actually imply a great deal about the properties of the natural number system. We will now state and prove a great deal of results about the successor function and the addition and multiplication operators. Many of these results will give us practice in the fundamental skill of proof by induction, and we will use these results implicitly for the rest of this text.

Proposition 1.3.4 Associativity of addition. $(l + m) + n = l + (m + n)$ for all natural numbers $l, m, n \in \mathbb{N}$.

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$ to show that $(l + m) + n = l + (m + n)$.

Base case. If $n = 0$, then

$$(l + m) + n = (l + m) + 0 = l + m = l + (m + 0) = l + (m + n).$$

Inductive step. Now suppose for the inductive hypothesis that $(l + m) + n =$

$l + (m + n)$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned}(l + m) + (n + 1) &= ((l + m) + n) + 1 = (\ell + (m + n)) + 1 \\ &= \ell + ((m + n) + 1) = \ell + (m + (n + 1)).\end{aligned}$$

We conclude that $(l + m) + n = l + (m + n)$ for all natural numbers $n \in \mathbb{N}$. ■

One consequence of [Proposition 1.3.4 Associativity of addition](#) is that we may unambiguously parse expressions with multiple sums without the need of parentheses; for example, if $l, m, n \in \mathbb{N}$ are natural numbers, then the expression $l + m + n$ is well-defined, because

$$(l + m) + n = l + (m + n).$$

Going forward, we will make use of these expressions wherever it is possible to do so without obscuring the relevant arguments.

[Proposition 1.3.4 Associativity of addition](#) implies that the result of adding a finite number of natural numbers depends only on the order of the numbers from left to right, and not on the iterative structure of the summands induced by parentheses. In fact, the order of the summands does not matter either. In order to show this, it helps to notice the following two results about addition.

Lemma 1.3.5 Addition by 0. $n + 0 = 0 + n = n$ for all natural numbers $n \in \mathbb{N}$.

Proof. $n + 0 = n$ for all natural numbers $n \in \mathbb{N}$ by the Peano axioms, and so it suffices to show $0 + n = n$. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$0 + n = 0 + 0 = 0 = n.$$

Inductive step. Now suppose for the inductive hypothesis that $0 + n = n$ for some natural number $n \in \mathbb{N}$, and note that

$$0 + (n + 1) = (0 + n) + 1 = n + 1.$$

We conclude that $0 + n = n$ for all natural numbers $n \in \mathbb{N}$. ■

Lemma 1.3.6 Towards commutativity of addition. $1 + n = n + 1$ for all natural numbers $n \in \mathbb{N}$.

Proof. By induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$1 + n = 1 + 0 = 0 + 1 = n + 1$$

by [Lemma 1.3.5 Addition by 0](#).

Inductive step. Now suppose for the inductive hypothesis that $1 + n = n + 1$ for some natural number $n \in \mathbb{N}$, and note that

$$1 + (n + 1) = (1 + n) + 1 = (n + 1) + 1$$

by [Proposition 1.3.4 Associativity of addition](#). We conclude that $1 + n = n + 1$ for all natural numbers $n \in \mathbb{N}$. ■

Proposition 1.3.7 Commutativity of addition. $m + n = n + m$ for all natural numbers $m, n \in \mathbb{N}$.

Proof. Let $m \in \mathbb{N}$ be a natural number. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$m + n = m + 0 = m = 0 + m = n + m$$

by [Lemma 1.3.5 Addition by 0](#).

Inductive step. Now suppose for the inductive hypothesis that $m + n = n + m$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned} m + (n + 1) &= (m + n) + 1 = (n + m) + 1 = 1 + (n + m) \\ &= (1 + n) + m = (n + 1) + m \end{aligned}$$

by [Proposition 1.3.4 Associativity of addition](#) and [Lemma 1.3.6 Towards commutativity of addition](#). We conclude that $m + n = n + m$ for all natural numbers $n \in \mathbb{N}$. ■

[Proposition 1.3.4 Associativity of addition](#) and [Proposition 1.3.7 Commutativity of addition](#) together imply that, when adding natural numbers, neither the order of the summands nor the order of the addition operations performed matters. In order to see that analogous properties also hold for multiplication, we will require the following supporting results. Along the way, we will also prove the distributivity of multiplication over addition.

Lemma 1.3.8 Multiplication by 1. $n \cdot 1 = n$ and $1n = n$ for all natural numbers $n \in \mathbb{N}$.

Proof. For all natural numbers $n \in \mathbb{N}$,

$$n \cdot 1 = n(0 + 1) = (n \cdot 0) + n = 0 + n = n$$

by [Lemma 1.3.5 Addition by 0](#). Thus it suffices to prove that $1n = n$ for all natural numbers $n \in \mathbb{N}$. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$1n = 1 \cdot 0 = 0 = n.$$

Inductive step. Now suppose for the inductive hypothesis that $1n = n$ for some natural number $n \in \mathbb{N}$, and note that

$$1(n + 1) = 1n + 1 = n + 1.$$

We conclude that $1n = n$ for all natural numbers $n \in \mathbb{N}$. ■

Proposition 1.3.9 Left distributivity of multiplication over addition. $l(m + n) = lm + ln$ for all natural numbers $l, m, n \in \mathbb{N}$.

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$l(m + n) = l(m + 0) = lm = lm + 0 = lm + l \cdot 0 = lm + ln.$$

Inductive step. Now suppose for the inductive hypothesis that $l(m + n) = lm + ln$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned} l(m + (n + 1)) &= l((m + n) + 1) = l(m + n) + l \\ &= (lm + ln) + l = lm + (ln + l) = lm + l(n + 1) \end{aligned}$$

by [Proposition 1.3.4 Associativity of addition](#). We conclude that $l(m+n) = lm + ln$ for all natural numbers $n \in \mathbb{N}$. ■

Proposition 1.3.10 Associativity of multiplication. $(lm)n = l(mn)$ for all natural numbers $l, m, n \in \mathbb{N}$.

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$(lm)n = (lm)0 = 0 = l \cdot 0 = l(m \cdot 0) = l(mn).$$

Inductive step. Now suppose for the inductive hypothesis that $(lm)n = l(mn)$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned} (lm)(n+1) &= (lm)n + lm = l(mn) + lm = \\ &= l(mn + m) = l(m(n+1)) \end{aligned}$$

by [Proposition 1.3.9 Left distributivity of multiplication over addition](#). We conclude that $(lm)n = l(mn)$ for all natural numbers $n \in \mathbb{N}$. ■

Just as with addition, one consequence of [Proposition 1.3.10 Associativity of multiplication](#) is that we may unambiguously parse expressions with multiple products without the need of parentheses; for example, if $l, m, n \in \mathbb{N}$ are natural numbers, then the expression lmn is well-defined, because

$$(lm)n = l(mn).$$

Going forward, we will make use of these expressions wherever it is possible to do so without obscuring the relevant arguments.

Proposition 1.3.11 Right distributivity of multiplication over addition. $(l+m)n = ln + mn$ for all natural numbers $l, m, n \in \mathbb{N}$.

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$(l+m)n = (l+m) \cdot 0 = 0 = 0 + 0 = l \cdot 0 + m \cdot 0 = ln + mn.$$

Inductive step. Now suppose for the inductive hypothesis that $(l+m)n = ln + mn$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned} (l+m)(n+1) &= (l+m)n + (l+m) = ln + l + mn + m \\ &= l(n+1) + m(n+1) \end{aligned}$$

by repeated applications of [Proposition 1.3.4 Associativity of addition](#) and [Proposition 1.3.7 Commutativity of addition](#). We conclude that $(l+m)n = ln + mn$ for all natural numbers $n \in \mathbb{N}$. ■

Lemma 1.3.12 Multiplication by 0. $0n = 0$ for all natural numbers $n \in \mathbb{N}$.

Proof. By induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$0n = 0 \cdot 0 = 0.$$

Inductive step. Now suppose for the inductive hypothesis that $0n = 0$ for some natural number $n \in \mathbb{N}$, and note that

$$0(n+1) = 0n + 0 = 0 + 0 = 0.$$

We conclude that $0n = 0$ for all natural numbers $n \in \mathbb{N}$. ■

Proposition 1.3.13 Commutativity of multiplication. $mn = nm$ for all natural numbers $m, n \in \mathbb{N}$.

Proof. Let $m \in \mathbb{N}$ be a natural number. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$mn = m \cdot 0 = 0 = 0 \cdot m = nm$$

by [Lemma 1.3.12 Multiplication by 0](#).

Inductive step. Now suppose for the inductive hypothesis that $mn = nm$ for some natural number $n \in \mathbb{N}$, and note that

$$m(n+1) = mn + m = nm + 1m = (n+1)m$$

by [Lemma 1.3.8 Multiplication by 1](#) We conclude that $mn = nm$ for all natural numbers $n \in \mathbb{N}$. ■

Lemma 1.3.14 Cancellation of addition. For all natural numbers $n \in \mathbb{N}$, addition by n is injective; that is, for all natural numbers $l, m, n \in \mathbb{N}$, if $l + n = m + n$, then $l = m$.

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$ and $l + n = m + n$, then

$$l = l + 0 = l + n = m + n = m + 0 = m$$

Inductive step. Now suppose for the inductive hypothesis that there is some natural number $n \in \mathbb{N}$ so that if $l + n = m + n$, then $l = m$. Note that if $l + (n+1) = m + (n+1)$, then

$$(l+n)+1 = l+(n+1) = m+(n+1) = (m+n)+1,$$

and so $l+n = m+n$ by the injectivity of the successor function. Thus $l = m$ by the inductive hypothesis. In summary, we have shown that if $l + (n+1) = m + (n+1)$, then $l = m$. We conclude that for all natural numbers $n \in \mathbb{N}$, if $l + n = m + n$, then $l = m$. ■

A similar cancellation law exists for multiplication, but its proof will require some properties of the comparison of natural numbers via its standard ordering.

1.3.3 Properties of Comparison

The final component of Peano's axiomatization of the natural numbers $\mathbb{N}$ is that of comparison. The standard ordering $\leq$ on the natural numbers has several important and familiar properties which we will now derive from first principles.

Proposition 1.3.15 Comparison is a total order. Comparison of natural numbers has the following properties:

1. Reflexivity.

$$n \leq n \text{ for all natural numbers } n \in \mathbb{N}.$$

2. Antisymmetry.

For all natural numbers $m, n \in \mathbb{N}$, if $m \leq n$ and $n \geq m$, then $m = n$.

3. *Transitivity.*

For all natural numbers $l, m, n \in \mathbb{N}$, if $l \leq m$ and $m \leq n$, then $l \leq n$.

4. *Totality.*

For all natural numbers $m, n \in \mathbb{N}$, either $m \leq n$ or $n \leq m$.

Proof.

1. For all natural numbers $n \in \mathbb{N}$, $n + 0 = n$, so that $n \leq n$.
2. Let $m, n \in \mathbb{N}$ be natural numbers, and suppose that $m \leq n$ and $n \leq m$. Then $n = m + p$ and $m = n + q$ for some natural numbers $p, q \in \mathbb{N}$. Note that

$$m + 0 = m = n + q = m + p + q,$$

and so $0 = p + q$ by [Lemma 1.3.14 Cancellation of addition](#); that is, $q \leq 0$. [Lemma 1.3.2 0 is minimal](#) now implies that $q = 0$, so that

$$m = n + q = n + 0 = n.$$

3. Let $l, m, n \in \mathbb{N}$ be natural numbers, and suppose that $l \leq m$ and $m \leq n$. Then $l + p = m$ and $m + q = n$ for some natural numbers $p, q \in \mathbb{N}$, and so

$$l + (p + q) = (l + p) + q = m + q = n.$$

We conclude that $l \leq n$.

4. Let $m \in \mathbb{N}$ be a natural number. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then $0 + m = m$ by [Lemma 1.3.5 Addition by 0](#), and so $n = 0 \leq m$.

Inductive step. Now suppose for the inductive hypothesis that there is a natural number $n \in \mathbb{N}$ so that either $m \leq n$ or $n \leq m$. If $m \leq n$, then $m + p = n$ for some natural number $p \in \mathbb{N}$, and so

$$m + (p + 1) = (m + p) + 1 = n + 1;$$

that is, $m \leq n + 1$.

Now suppose that $n \leq m$, so that $n + p = m$ for some natural number $p \in \mathbb{N}$. If $p = 0$, then

$$m + 1 = (n + p) + 1 = (n + 0) + 1 = n + 1,$$

and so $m \leq n + 1$. On the other hand, if $p \neq 0$, then $p = q + 1$ for some natural number $q \in \mathbb{N}$ by an argument in the proof of [Lemma 1.3.2 0 is minimal](#). Then

$$(n + 1) + q = n + 1 + q = n + q + 1 = n + p = m,$$

and so $n + 1 \leq m$. We conclude that for all natural numbers $n \in \mathbb{N}$, either $m \leq n$ or $n \leq m$.

■

Corollary 1.3.16 Cancellation of multiplication. *For all natural numbers $n \in \mathbb{N}$, if $n \neq 0$, then addition by n is injective; that is, for all natural numbers $l, m, n \in \mathbb{N}$, if $ln = mn$, then either $l = m$ or $n = 0$.*

Proof. By induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then $n = 0$, and so the desired conclusion holds for all natural numbers $l, m \in \mathbb{N}$.

Inductive step. Now suppose for the inductive hypothesis that there is a natural number $n \in \mathbb{N}$ so that for all natural numbers $l, m \in \mathbb{N}$, if $ln = mn$, then either $n = 0$ or $l = m$. Let $l, m \in \mathbb{N}$ be natural numbers, and suppose that $l(n+1) = m(n+1)$. Then $n+1 \neq 0$, and so it suffices to show that $l = m$.

To that end, note that (4) of [Proposition 1.3.15 Comparison is a total order](#) implies that either $l \leq m$ or $m \leq l$. Without loss of generality, we may suppose that $l \leq m$, so that $m = l + p$ for some natural number $p \in \mathbb{N}$. We observe that

$$\begin{aligned} 0 + (ln + l) &= ln + l = l(n+1) = m(n+1) \\ &= (l+p)(n+1) = ln + l + pn + p = (pn + p) + (ln + l) \end{aligned}$$

by [Lemma 1.3.5 Addition by 0](#), [Proposition 1.3.9 Left distributivity of multiplication over addition](#), [Proposition 1.3.11 Right distributivity of multiplication over addition](#), [Proposition 1.3.4 Associativity of addition](#), and [Proposition 1.3.7 Commutativity of addition](#).

In particular, [Lemma 1.3.14 Cancellation of addition](#) implies that $0 = pn + p$, so that $p \leq 0$. We conclude by [Lemma 1.3.2 0 is minimal](#) that $p = 0$, so that

$$l = m + p = m + 0 = m.$$

We conclude that for all natural numbers $l, m, n \in \mathbb{N}$, either if $ln = mn$, then either $l = m$ or $n = 0$. ■

Lemma 1.3.17 Comparison and arithmetic. *Let $l, m, n, p \in \mathbb{N}$ be natural numbers.*

1. *If $l \leq n$ and $m \leq p$, then $l + m \leq n + p$.*
2. *If $l \leq n$ and $m \leq p$, then $lm \leq np$.*

Proof.

1. If $l \leq n$ and $m \leq p$, then $n = l + q$ and $p = m + r$ for some natural numbers $q, r \in \mathbb{N}$, and so

$$(l + m) + (q + r) = l + m + q + r = l + q + m + r = n + p.$$

We conclude that $l + m \leq n + p$.

2. If $l \leq n$ and $m \leq p$, then $n = l + q$ and $p = m + r$ for some natural numbers $q, r \in \mathbb{N}$, and so

$$\begin{aligned} lm + (lr + qm + qr) &= lm + lr + qm + qr = l(m + r) + q(m + r) \\ &= (l + q)(m + r) = np. \end{aligned}$$

We conclude that $lm \leq np$. ■

For ease of reference, we collect the above results (along with some imme-

diate corollaries) in the following theorem.

Theorem 1.3.18 Properties of Peano arithmetic. *Addition, multiplication, and comparison of natural numbers have the following properties:*

1. *Associativity of addition.*

$$(l + m) + n = l + (m + n) \text{ for all natural numbers } l, m, n \in \mathbb{N}.$$

2. *Commutativity of addition.*

$$m + n = n + m \text{ for all natural numbers } m, n \in \mathbb{N}.$$

3. *Additive identity.*

$$n + 0 = 0 + n = n \text{ for all natural numbers } n \in \mathbb{N}.$$

4. *Associativity of multiplication.*

$$(lm)n = l(mn) \text{ for all natural numbers } l, m, n \in \mathbb{N}.$$

5. *Commutativity of multiplication.*

$$mn = nm \text{ for all natural numbers } m, n \in \mathbb{N}.$$

6. *Multiplicative annihilation.*

$$n \cdot 0 = 0n = 0 \text{ for all natural numbers } n \in \mathbb{N}.$$

7. *Multiplicative identity.*

$$n \cdot 1 = 1n = n \text{ for all natural numbers } n \in \mathbb{N}.$$

8. *Distributivity of multiplication over addition.*

$$l(m + n) = lm + ln \text{ and } (l + m)n = ln + mn \text{ for all natural numbers } l, m, n \in \mathbb{N}.$$

9. *Reflexivity of comparison.*

$$n \leq n \text{ for all natural numbers } n \in \mathbb{N}.$$

10. *Antisymmetry of comparison.*

$$\text{For all natural numbers } m, n \in \mathbb{N}, \text{ if } m \leq n \text{ and } n \leq m, \text{ then } m = n.$$

11. *Transitivity of comparison.*

$$\text{For all natural numbers } l, m, n \in \mathbb{N}, \text{ if } l \leq m \text{ and } m \leq n, \text{ then } l \leq n.$$

12. *Totality of comparison.*

$$\text{For all natural numbers } m, n \in \mathbb{N}, \text{ either } m \leq n \text{ or } n \leq m.$$

13. *Compatibility of comparison and addition/multiplication.*

$$\text{For all natural numbers } l, m, n, p \in \mathbb{N}, \text{ if } l \leq n \text{ and } m \leq p, \text{ then}$$

$$l + m \leq n + p$$

and

$$lm \leq np$$

Many of these properties of the arithmetic and comparison of natural numbers also extend to the arithmetic of the other classes of numbers we will consider, namely the integers, rational numbers, real numbers, and complex numbers.

In the next sections, we will generalize the properties of addition and multiplication and the order-theoretic properties of comparison of natural numbers

and arrive at fundamental notions of binar operations and relations, partial and total orders, and the well-ordering principle.

1.3.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

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In-depth solutions to these reading questions are available. However, these are provided in order to check your work. If you are unable to solve the above reading questions, discuss the problem with classmates and/or instructor rather than spoiling the answers.

1.4 Finitary Operations and Relations

In this section, we will develop the notion of (finitary) operations and relations on sets. Intuitively, a **finitary operation** on a set is an assignment of an element of that set to some finite number of other elements, and a **finitary relation** on a set is a designation of certain groups of some finite number of elements as associated.

1.4.1 Products and Choice

Previously, we have used the mathematical convention of indexing to generalize the intersection and union of two sets. We may also use indexing to generalize the Cartesian product of two sets to act on arbitrary indexed families of sets. In this section, we will develop this generalized Cartesian product, and use it to introduce operations and relations on a set. Instead of containing ordered pairs of elements, our new Cartesian product consists of indexed families of elements.

Definition 1.4.1 Cartesian product. Let $(U_j)_{j \in \mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X , called **factors**, indexed by a set $\mathcal{J}$. The **Cartesian**

product $\prod_{j \in \mathcal{J}} U_j$ of this indexed family is the set

$$\prod_{j \in \mathcal{J}} U_j = \left\{ (u_j)_{j \in \mathcal{J}} \in X^{\mathcal{J}} \mid u_j \in U_j \text{ for all } j \in \mathcal{J} \right\}$$

of families $(u_j)_{j \in \mathcal{J}}$ of elements $u_j \in X$ with the property that $u_j \in U_j$ for all indices $j \in \mathcal{J}$.

For each index $k \in \mathcal{J}$, the k th **projection map** $\pi_{\prod_{j \in \mathcal{J}} U_j}^{U_k} : \prod_{j \in \mathcal{J}} U_j \rightarrow U_k$ is the map from the Cartesian product $\prod_{j \in \mathcal{J}} U_j$ to the k th factor U_k defined by the formula

$$\pi_{\prod_{j \in \mathcal{J}} U_j}^{U_k} (u_j)_{j \in \mathcal{J}} = u_k.$$

◆

Convention 1.4.2 Empty and singleton products. What does a family indexed by the empty set look like? The empty set $\emptyset: \emptyset \rightarrow X$ is the unique map from the empty set $\emptyset$ to a set X . Therefore, there is only one such family

$$(x_j)_{j \in \emptyset} = \emptyset$$

of elements x_j of X indexed by the empty set $\emptyset$. This unique indexed family is usually denoted $() = \emptyset$. By the above remark,

$$\prod_{j \in \emptyset} U_j = \{()\} = \{\emptyset\}.$$

Similarly, a family $(x_j)_{j \in \{j\}}$ of elements $x_j \in X$ of a set S indexed by a singleton set $\{j\}$ is uniquely determined by the image x_j of j . Such an indexed family is denoted

$$(x_j) = (x_j)_{j \in \{j\}}.$$

We usually act under the convention $(x_j) = x_j$, so that

$$\prod_{j \in \{j\}} U_j = U_j.$$

It may seem intuitive or obvious that the Cartesian product of an indexed family of nonempty sets is itself nonempty. After all, one can specify an indexed family of elements of these sets just by choosing an element of each nonempty set. This notion of a choice of elements of nonempty sets gives the following axiom its name.

Axiom 1.4.3 Axiom of choice. *Let $(U_j)_{j \in \mathcal{J}}$ be an indexed family of subsets $U_j \subseteq X$ of a set X . If $U_j \neq \emptyset$ for all indices $j \in \mathcal{J}$, then the Cartesian product $\prod_{j \in \mathcal{J}} U_j \neq \emptyset$ is also non-empty.*

We will take the [Axiom of choice](#) as true for the remainder of this text. While seemingly too specific to be of much use, the axiom of choice can be used to prove many useful results. In particular, we can use it to complete the characterization of invertibility given in [Theorem 1.2.15 Invertibility](#).

Corollary 1.4.4 Invertibility. *Let $f: X \rightarrow Y$ be a map from a set X to a set Y .*

1. *f is left-invertible if and only if f is injective or $X = \emptyset$ is empty.*
2. *f is right-invertible if and only if f is surjective.*
3. *f is invertible if and only if f is bijective.*

Proof.

1. This follows directly from (1) and (2) of [Theorem 1.2.15 Invertibility](#).
2. The forward implication is (3) of [Theorem 1.2.15 Invertibility](#). Thus we may suppose that f is surjective. Then for each output $y \in Y$, the pre-image

$$f^{-1}(y) = \{x \in X \mid f(x) = y\} \neq \emptyset$$

is a non-empty subset of X , and so [Axiom 1.4.3 Axiom of choice](#) implies that the Cartesian product

$$\prod_{y \in Y} f^{-1}(y) \neq \emptyset$$

is a non-empty subset of X^Y . Choose an element $g = (x_y)_{y \in Y} \in \prod_{y \in Y} f^{-1}(y)$, and note that for all $y \in Y$, $g(y) = x_y \in f^{-1}(y)$, so that

$$(f \circ g)(y) = f(g(y)) = y = \text{id}_Y(y);$$

that is, $f \circ g = \text{id}_Y$ so that g is a right inverse to f . Hence f is right-invertible.

3. This is (3) of [Theorem 1.2.15 Invertibility](#). ■

Definition 1.4.5 Cartesian power. Let X be a set and $m \in \mathbb{N}$ be a natural number. The m th **Cartesian power** X^m of X is the Cartesian product

$$X^m = \prod_{j=1}^m X = \left\{ (x_j)_{j=1}^m \mid x_j \in X \text{ for all } 1 \leq j \leq m \right\}.$$

◆

We will now move on to study the Cartesian powers of sets. Maps from the Cartesian power of a set to the set itself will be called finitary operations on that set, and subsets of the Cartesian power of a set will be called finitary relations on that set.

1.4.2 Finitary Operations

Informally, a finitary operation on a set is an assignment of its elements to all finite sequence of a certain length in that set. That is, a finitary operation on a set is a map from some finite Cartesian power of that set to the set itself. We will view a finitary operation (through the active characterization of what maps do) as a process which combines elements of a set into a new element.

Definition 1.4.6 Finitary operation. Let X be a set and $n \in \mathbb{N}$ be a natural number. A **finitary operation** of **arity** n (or more simply a **n -ary operation**) is a map $f: X^n \rightarrow X$. ◆

Example 1.4.7 Finitary operations. The following are examples of finitary operations:

1. A 0-ary operation (or more simply, a **nullary operation**) on a set X is just a map $f: \{\emptyset\} \rightarrow X$ from $X^0 = \{\emptyset\}$ to X . Such a map $f: \{\emptyset\} \rightarrow X$ is uniquely determined by the image $f(\emptyset) \in X$ of the unique element $\emptyset \in X^0$. In this way, we may view the nullary operations on X as

synonymous with the elements of X .

2. A 1-ary operation (or more simply, a **unary operation**) on a set X is just a map $f: X \rightarrow X$. One example of such a unary operation on the natural numbers $\mathbb{N}$ is the successor function $\text{succ}: \mathbb{N} \rightarrow \mathbb{N}$.
3. A 2-ary operation (or more simply, a **binary operation**) on a set X is a map $f: X^2 \rightarrow X$. Two examples of such binary operations on the natural numbers $\mathbb{N}$ are the addition and multiplication operations $\text{add}, \text{mult}: \mathbb{N}^2 \rightarrow \mathbb{N}$.

Binary operations are usually denoted with symbols such as $+$, $\times$, $\cdot$, $*$, and $\star$. For a given binary operation $*$: $X \times X \rightarrow X$, we write

$$x_1 * x_2 = *(x_1, x_2)$$

for all elements $x_1, x_2 \in X$.



In many ways, algebra is the study of finitary operations on sets. Particularly emphasized in group theory and ring theory is the study of binary operation on sets. We won't delve too deep into the study of binary operations, but we will introduce some nice properties which some binary operations may satisfy.

Definition 1.4.8 Commutativity; associativity. Let $*$: $X^2 \rightarrow X$ be a binary operation on a set X .

1. $*$ is called **commutative** if

$$x_1 * x_2 = x_2 * x_1$$

for all elements $x_1, x_2 \in X$.

2. $*$ is called **associative** if

$$(x_1 * x_2) * x_3 = x_1 * (x_2 * x_3)$$

for all elements $x_1, x_2, x_3 \in X$.



Example 1.4.9 Commutative/associative binary operations. The following are examples of commutativity and associativity in binary operations:

1. Addition $+: \mathbb{N}^2 \rightarrow \mathbb{N}$ and multiplication $\cdot: \mathbb{N}^2 \rightarrow \mathbb{N}$ are commutative and associative binary operations on $\mathbb{N}$ by (1), (2), (4), and (5) of [Theorem 1.3.18 Properties of Peano arithmetic](#).
2. Intersection $\cap: \mathcal{P}(X)^2 \rightarrow \mathcal{P}(X)$ and union $\cup: \mathcal{P}(X)^2 \rightarrow \mathcal{P}(X)$ are commutative and associative binary operations on the power set $\mathcal{P}(X)$ of a set X by (2) and (3) of [Corollary 1.1.15 Properties of the set operations](#).
3. Composition $\circ: (X^X)^2 \rightarrow X^X$ is an associative but non-commutative binary operation on the set X^X of maps from a set X to itself.
4. Let $e: \mathbb{N}^2 \rightarrow \mathbb{N}$ be the **exponentiation** function defined by the formula $e(m, n) = m^n$; that is, e is defined recursively by the formulae $e(m, 0) = 1$ and $e(m, n + 1) = e^{m, n} \cdot m$. Then e is a binary operation on $\mathbb{N}$, but e is neither commutative nor associative.



Associative binary operations can be extended recursively to act on finite sequences of any length. For example, given a finite sequence $(n_j)_{j=1}^m$ of natural numbers $n_j \in \mathbb{N}$, we write

$$\sum_{j=1}^m n_j = n_1 + n_2 + \cdots + n_m \qquad \prod_{j=1}^m n_j = n_1 n_2 \cdots n_m.$$

As in the case of finitary operations, we will introduce finitary relations of all arities, but we will emphasize binary relations in order to study congruences and order relations, which are ubiquitous throughout both classical and modern mathematics and which will recur throughout this text.

1.4.3 Finitary Relations

A finitary relation on a set is just a subset of some finite Cartesian power of the set. This subset is usually interpreted as distinguishing its elements certain finite sequences of elements of the set as important in some way.

Definition 1.4.10 Finitary relation. Let X be a set and $n \in \mathbb{N}$ be a natural number. A **finitary relation** of **arity** n (or more simply a **n -ary relation**) is a subset $R \subseteq X^n$. ◆

We will view a finitary relation on a set as elevating specific finite sequences of elements of that set as distinguished.

Example 1.4.11 Finitary relations. The following are examples of finitary operations:

1. In light of the convention $X^1 = X$, a 1-ary operation (or more simply, a **unary operation**) on a set X is just a subset of X .
2. A 2-ary relation (or more simply, a **binary relation**) on a set X is a subset $R \subseteq X^2$. Two examples of such binary relations are the containment relation $\subseteq$ on the power set $\mathcal{P}(X)$ of a set X and the standard ordering $\text{comp} \subseteq \mathbb{N}^2$ of the natural numbers $\mathbb{N}$.

Just as there is a special notation for binary operations, there is a special relation for binary relations as well. Specifically, for a given binary relation $R \subseteq X^2$ on a set X , we write $x_1 R x_2$ to mean that $(x_1, x_2) \in R$. ▲

Definition 1.4.12 Pullback of a finitary relation. Let X and Y be sets and $n \in \mathbb{N}$ be a natural number. The **pullback** R_f of an n -ary relation R on Y along a map $f: X \rightarrow Y$ is the n -ary relation on X defined by

$$R_f = \{(x_k)_{k=1}^n \in X^n \mid (f(x_k))_{k=1}^n \in R\}.$$

Example 1.4.13 Restriction of a finitary relation. Let X be a set and $n \in \mathbb{N}$ be a natural number. The **restriction** $R|_U$ of an n -ary relation R on X to a subset $U \subseteq X$ is the pullback of R along the inclusion map $\iota_U^X: U \rightarrow X$; that is, ◆

$$\begin{aligned} R|_U &= R_{\iota_U^X} = \left\{ (u_k)_{k=1}^n \in U^n \mid (\iota_U^X(u_k))_{k=1}^n \in R \right\} = R \cap U^n \\ &= \{(x_k)_{k=1}^n \in X^n \mid x_1, \dots, x_n \in U \text{ and } (x_k)_{k=1}^n \in R\} = R \cap U^n. \end{aligned}$$

▲

One particularly nice type of binary relation is the **equivalence relation**. This is a generalization of the notion of equality, which has many of the salient properties of the equality relation. We will see that an equivalence relation partitions a set into disjoint subsets known as **equivalence classes**. When we consider more sophisticated structures than just sets, we will use equivalence relations to construct quotient objects, as is done in algebraic settings such as group theory or ring theory.

Definition 1.4.14 Equivalence relation. Let $\equiv$ be a binary relation on a set X .

1. $\equiv$ is called **reflexive** if $x \equiv x$ for all elements $x \in X$.
2. $\equiv$ is called **symmetric** if for all elements $x_1, x_2 \in X$, if $x_1 \equiv x_2$, then $x_2 \equiv x_1$.
3. $\equiv$ is called **transitive** if for all elements $x_1, x_2, x_3 \in X$, if $x_1 \equiv x_2$ and $x_2 \equiv x_3$, then $x_1 \equiv x_3$.

$\equiv$ is called an **equivalence relation** if it is reflexive, symmetric, and transitive. ◆

Lemma 1.4.15 Pullbacks of equivalence relations. *The pullback $\equiv_f$ of an equivalence relation $\equiv$ on a set Y along a map $f: X \rightarrow Y$ from a set X to Y is an equivalence relation on X .*

Proof. First let $x \in X$, and note that $f(x) \equiv f(x)$ by the reflexivity of $\equiv$, and so $x \equiv_f x$; that is, $\equiv_f$ is reflexive.

Now let $x_1, x_2 \in X$, and suppose that $x_1 \equiv_f x_2$. Then $f(x_1) \equiv f(x_2)$, and so $f(x_2) \equiv f(x_1)$ by the symmetry of $\equiv$. In particular, $x_2 \equiv_f x_1$, and so $\equiv_f$ is symmetric.

Finally, let $x_1, x_2, x_3 \in X$, and suppose that $x_1 \equiv_f x_2$ and $x_2 \equiv_f x_3$. Then $f(x_1) \equiv f(x_2)$ and $f(x_2) \equiv f(x_3)$, and so $f(x_1) \equiv f(x_3)$ by the transitivity of $\equiv$. In particular, $x_1 \equiv_f x_3$, and so $\equiv_f$ is transitive.

Since $\equiv_f$ is reflexive, symmetric, and transitive, it is an equivalence relation on X . ■

Definition 1.4.16 Equivalence class; quotient set. Let $\equiv$ be an equivalence relation on a set X . The **equivalence class** $[x]_{\equiv}$ of an element $x \in X$ is the subset of X defined by

$$[x]_{\equiv} = \{y \in X \mid x \equiv y\};$$

that is, $[x]_{\equiv}$ is the set of elements which are equivalent to x under the relation $\equiv$. The **quotient** $X/\equiv$ of X by $\equiv$ is the set of equivalence classes; that is,

$$X/\equiv = \{[x]_{\equiv} \in \mathcal{P}(X) \mid x \in X\}.$$

As mentioned above, an equivalence relation on a given set partitions that set into pairwise disjoint equivalence classes. ◆

Proposition 1.4.17 Equivalence partition. *The equivalence classes of an equivalence relation $\equiv$ on a set X form a **partition** of X ; that is, two equivalence classes are either equal or disjoint, and*

$$X = \bigcup_{c \in X/\equiv} c.$$

Proof. Each equivalence class is a subset of X , and so

$$\bigcup_{c \in X/\equiv} c \subseteq X.$$

In order to show the reverse containment, let $x \in X$, and note that $x \equiv x$ by reflexivity; that is, $x \in [x]_{\equiv}$, and so

$$x \in [x]_{\equiv} = \bigcup_{c \in X/\equiv} c.$$

We conclude that

$$X = \bigcup_{c \in X/\equiv} c.$$

It remains to show that two equivalence classes are either equal or disjoint. To that end, consider elements $x_1, x_2 \in X$. First suppose that $x_1 \equiv x_2$, and let $x \in [x_2]_{\equiv}$. Then $x_2 \equiv x$, and so transitivity implies that $x_1 \equiv x$; that is, $x \in [x_1]_{\equiv}$.

In summary, we have shown that if $x_1 \equiv x_2$, then $[x_2]_{\equiv} \subseteq [x_1]_{\equiv}$. Symmetry implies that if $x_1 \equiv x_2$, then $x_2 \equiv x_1$, and so $[x_1]_{\equiv} \subseteq [x_2]_{\equiv}$; that is, if $x_1 \equiv x_2$, then $[x_1]_{\equiv} = [x_2]_{\equiv}$.

Conversely, now suppose that the equivalence classes $[x_1]_{\equiv}$ and $[x_2]_{\equiv}$ are not disjoint, so that $x \in [x_1]_{\equiv} \cap [x_2]_{\equiv}$ for some element $x \in X$. Since $x \in [x_1]_{\equiv}$ and $x \in [x_2]_{\equiv}$, $x \equiv x_1$ and $x \equiv x_2$. We conclude by symmetry and transitivity that $x_1 \equiv x_2$. ■

Equivalence relations are ubiquitous throughout mathematics, and we will work with them for the remainder of this text.

While this section has dealt with finitary operations and relations in broad generality, the vast majority of mathematics relevant to this text deals with binary operations and relations. In the next section, we will see another use for binary relations; that of ordering sets.

1.4.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

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1.5 Introduction to Order Theory

In this section, we will investigate another class of binary relations, which can be used to establish a notion of an ordering on a set. Touchstone examples will be the standard ordering of the natural numbers and the ordering of sets by containment.

1.5.1 Partial Orders

We have seen that a finitary relation on a set can be viewed as elevating its elements as distinguished finite sequences in that set. Some binary relations on a set may be interpreted in this way as establishing an ordering, with the first entries of each distinguished ordered pair preceding the second entries in this ordering.

What are the salient properties of the standard ordering of the natural numbers? We will see that, in order to establish a similar notion of ordering on a set, a given binary operation should satisfy similar properties. However, even if a binary operation satisfies some, but not all, of the desired properties, we will see that we can still interpret it as a weaker type of ordering relation. However, all ordering relations should, at the very least, satisfy reflexivity and transitivity. Such a reflexive and transitive binary operation is called a preorder.

Definition 1.5.1 Preorder. A binary relation $\preceq$ on a set X is called a **preorder** if it is reflexive and transitive; that is, $\preceq$ is a preorder if it satisfies the following conditions:

1. *Reflexivity.*

$$x \preceq x \text{ for all elements } x \in X.$$

2. *Transitivity.*

$$\text{For all elements } x_1, x_2, x_3 \in X, \text{ if } x_1 \preceq x_2 \text{ and } x_2 \preceq x_3, \text{ then } x_1 \preceq x_3.$$

If $\preceq$ is a preorder on a set X , then the ordered pair $(X, \preceq)$ is called a **pre-ordered set** (or a **proset**). ♦

Lemma 1.5.2 Preorders from pullbacks. *The pullback $\leq_f$ of the standard ordering $\leq$ on the natural numbers $\mathbb{N}$ along a function $f: X \rightarrow \mathbb{N}$ on a set X is a preorder on X .*

Proof. First let $x \in X$, and note that $f(x) \leq f(x)$ by (1) of [Proposition 1.3.15 Comparison is a total order](#). In particular, $x \leq_f x$, and so $\leq_f$ is reflexive.

Now let $x_1, x_2, x_3 \in X$, and suppose that $x_1 \leq_f x_2$ and $x_2 \leq_f x_3$. Then $f(x_1) \leq f(x_2)$ and $f(x_2) \leq f(x_3)$, and so $f(x_1) \leq f(x_3)$ by (3) of [Proposition 1.3.15 Comparison is a total order](#). In particular, $x_1 \leq_f x_3$, and so $\leq_f$ is

transitive. Since $\leq_f$ is both reflexive and transitive, it is a preorder on X . ■

It turns out that, at least relative to the standard ordering $\leq$ of the natural numbers $\mathbb{N}$, the defining properties of a preorder are still too weak to be of much use in the general case. After all, an equivalence relation is just a symmetric preorder, and there is no way of meaningfully assigning a notion of an ordering based on an equivalence relation. This is because an equivalence relation is symmetric; in order to consider an ordering based on a binary operation, we would want to avoid any symmetry at all. Therefore, we must require that an ordering relation satisfy stronger assumptions.

Definition 1.5.3 Partial order. A binary relation $\preceq$ on a set X is called a **partial order** if it satisfies the following conditions:

1. *Reflexivity.*

$x \preceq x$ for all elements $x \in X$.

2. *Anti-symmetry.*

For all elements $x_1, x_2 \in X$, if $x_1 \preceq x_2$ and $x_2 \preceq x_1$, then $x_1 = x_2$.

3. *Transitivity.*

For all elements $x_1, x_2, x_3 \in X$, if $x_1 \preceq x_2$ and $x_2 \preceq x_3$, then $x_1 \preceq x_3$.

If $\preceq$ is a partial order on a set X , then the ordered pair $(X, \preceq)$ is called a **partially ordered set** (or a **poset**). ◆

Example 1.5.4 Partial orders. The following are examples of partial orders on sets:

1. The standard ordering $\leq$ on the natural numbers $\mathbb{N}$ is a partial order on $\mathbb{N}$ by (1), (2), and (3) of [Proposition 1.3.15 Comparison is a total order](#).
2. Containment $\subseteq$ is a partial order on the power set $\mathcal{P}(X)$ of a set X by [Definition 1.1.6 Set equality](#) and [Lemma 1.1.8 Transitivity of set containment](#).

▲

At the end of this section, we will explore a property of partial orders called **totality** which distinguishes between our touchstone examples of the standard ordering $\leq$ of natural numbers and the containment $\subseteq$ of sets.

1.5.2 Boundedness and Extrema

Partial orders satisfy many of the properties of the standard ordering of the natural numbers. These properties allow us to generalize the definitions of many of the usual notions associated with an ordering, such as upper and lower bounds and maximal and minimal elements, to partially ordered sets.

Definition 1.5.5 Boundedness. Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$.

1. An element $x \in X$ is called an **upper bound** for U if $u \preceq x$ for all elements $u \in U$. If such an upper bound for U exists, then U is called **bounded from above** in X . Otherwise, if no such upper bound for U exists, then U is called **unbounded from above**.
2. An element $x \in X$ is called a **lower bound** for U if $x \preceq u$ for all elements $u \in U$. If such a lower bound for U exists, then U is called **bounded from below** in X .

A subset $U \subseteq X$ is called **bounded** in X if it is both bounded from above in X and bounded from below in X , and U is called **unbounded** in X otherwise if it is either unbounded from above in X or unbounded from below in X . ♦

Lemma 1.5.6 Extrema are unique. *Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$. At most one element of U is an upper bound for U in X , and at most one element of U is a lower bound for U in X .*

Proof. Let $u_1, u_2 \in U$, and suppose first that u_1 and u_2 are upper bounds for U in X , so that $u \preceq u_1$ and $u \preceq u_2$ for all elements $u \in U$. In particular, $u_1 \preceq u_2$ and $u_2 \preceq u_1$, and so $u_1 = u_2$ by anti-symmetry.

Similarly, now suppose that u_1 and u_2 are lower bounds for U in X , so that $u_1 \preceq u$ and $u_2 \preceq u$ for all elements $u \in U$. In particular, $u_1 \preceq u_2$ and $u_2 \preceq u_1$, and so $u_1 = u_2$ by anti-symmetry. ■

Definition 1.5.7 Extrema. Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$.

1. If there is an element of U which is an upper bound for U in X , then this unique element is called the **maximum** of U and is denoted $\max(U)$.
2. If there is an element of U which is a lower bound for U in X , then this unique element is called the **minimum** of U and is denoted $\min(U)$.

Together, maxima and minima are called **extrema**. ♦

The uniqueness of extrema in partially ordered sets is one way in which partial orders resemble the standard ordering of the natural number system. However, unlike in the standard ordering of the natural number system, there is in general a distinction between extrema, which are defined above, and extremal elements, which are defined below.

Definition 1.5.8 Extremal element. Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$.

1. An element $u \in U$ is called **maximal** in U if for all elements $v \in U$, if $u \preceq v$, then $v = u$.
2. An element $u \in U$ is called **minimal** in U if for all elements $v \in U$, if $v \preceq u$, then $v = u$.

Together, maximal and minimal elements are called **extremal elements**. ♦

Example 1.5.9 Extrema and extremal elements. The following are examples of extrema and extremal elements:

1. The minimum of the natural numbers $\mathbb{N}$ is $\min(\mathbb{N}) = 0$, which is the unique minimal element. However, $\mathbb{N}$ has no maximum or maximal element.
2. Consider set containment $\subseteq$ on the power set $\mathcal{P}(X)$ of a set X . $\mathcal{P}(X)$ has minimum and maximum $\min(\mathcal{P}(X)) = \emptyset$ and $\max(\mathcal{P}(X)) = X$, which are also the unique minimal and maximal elements, respectively.
3. Now consider set containment $\subseteq$ on the set X of proper non-empty subsets of the set $\{1, 2, 3\}$; that is,

$$X = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}\}.$$

Then X has no maximum or minimum, but $\{1\}$, $\{2\}$, and $\{3\}$ are minimal elements, and $\{1, 2\}$, $\{1, 3\}$, and $\{2, 3\}$ are maximal elements. ▲

Proposition 1.5.10 Extrema are extremal elements. *Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$. If U has a maximum, then $\max(U)$ is the unique maximal element of U . If U has a minimum, then $\min(U)$ is the unique minimal element of U .*

Proof. First suppose that U has a maximum. Let $u \in U$, and suppose that $\max(U) \preceq u$. Of course $u \preceq \max(U)$, and so anti-symmetry implies that $u = \max(U)$; that is, $\max(U)$ is maximal in U .

On the other hand, now consider a maximal element $u \in U$. Since $u \preceq \max(U)$, the maximality of u implies that $u = \max(U)$.

Now suppose that U has a minimum. Let $u \in U$, and suppose that $u \preceq \min(U)$. Of course $\min(U) \preceq u$, and so anti-symmetry implies that $u = \min(U)$; that is, $\min(U)$ is minimal in U .

On the other hand, now consider a minimal element $u \in U$. Since $\min(U) \preceq u$, the minimality of u implies that $u = \min(U)$. ■

The distinction between extrema and extremal elements is one way in which more general partial orders differ from the standard ordering of the natural numbers. However, this subtlety does not arise in all partial orders. We will shortly explore a class of partial orders in which the aforementioned distinction between extrema and extremal elements does not exist.

1.5.3 Comparability and Total Orders

The fact that the converse of [Proposition 1.5.10 Extrema are extremal elements](#) does not hold in general stems from the fact that in a partially ordered set, not all pairs of elements are comparable. That is, given two elements of a partially ordered set, it is not necessarily true that one will precede the other.

Definition 1.5.11 Comparability. Two elements $x_1, x_2 \in X$ in a partially ordered set $(X, \preceq)$ are called **comparable** if either $x_1 \preceq x_2$ or $x_2 \preceq x_1$, and **incomparable** otherwise. ◆

Any two natural numbers are comparable in the standard ordering. However, this is not true for partially ordered sets in general. As an example, consider the subsets $S = \{0, 1\}$ and $T = \{1, 2\}$ of the natural numbers $\mathbb{N}$. One can check that $S \not\subseteq T$ and $T \not\subseteq S$, so that S and T are incomparable in the power set $\mathcal{P}(\mathbb{N})$ of the natural numbers $\mathbb{N}$. So partially ordered sets may have elements which are incomparable. This is the reason why partial orders are called partial; partial orders in which all elements are comparable are called **total**.

Definition 1.5.12 Total order. A binary relation $\leq$ on a set X is called a **total order** if it is anti-symmetric and transitive and each pair of elements is comparable; that is, $\leq$ is a total order if it satisfies the following conditions:

1. *Antisymmetry.*

For all elements $x_1, x_2 \in X$, if $x_1 \leq x_2$ and $x_2 \leq x_1$, then $x_1 = x_2$.

2. *Transitivity.*

For all elements $x_1, x_2, x_3 \in X$, if $x_1 \leq x_2$ and $x_2 \leq x_3$, then $x_1 \leq x_3$.

3. *Totality.*

For all elements $x_1, x_2 \in X$, either $x_1 \leq x_2$ or $x_2 \leq x_1$.

If $\leq$ is a total order on a set X , then the ordered pair $(X, \leq)$ is called a **totally ordered set**. ◆

Example 1.5.13 Total order. The standard ordering $\leq$ on the natural numbers $\mathbb{N}$ is a total order by (2), (3), and (4) of [Proposition 1.3.15 Comparison is a total order](#). ▲

Lemma 1.5.14 Total orders are partial orders. Every total order on a set X is a partial order on X .

Proof. Let $\leq$ be a total order on a set X . Since $\leq$ is antisymmetric and transitive, it remains to show that $\leq$ is reflexive. To that end, let $x \in X$, and note that totality implies that either $x \leq x$ or $x \leq x$; that is $\leq$ is reflexive (and hence a partial order on X). ■

Proposition 1.5.15 Extremal elements are extrema in totally ordered sets. Let $\preceq$ be a total order on a set X .

1. If X contains a maximal element $e \in X$, then $e = \max(X)$.
2. If X contains a minimal element $e \in X$, then $e = \min(X)$.

Proof.

1. Suppose that $e \in X$ is a maximal element, and let $x \in X$. Totality implies that $x \leq e$ or $e \leq x$. If $e \leq x$, then maximality implies that $x = e$, so that $x \leq e$ by reflexivity. Thus $e \leq x$ for all $x \in X$, and so $\max(X) = e$.
2. Now suppose that $e \in X$ is a minimal element, and let $x \in X$. Totality implies that $x \leq e$ or $e \leq x$. If $x \leq e$, then minimality implies that $x = e$, so that $e \leq x$ by reflexivity. Thus $x \leq e$ for all $x \in X$, and so $\min(X) = e$. ■

We end with an important property of the standard ordering $\leq$ on the natural numbers $\mathbb{N}$ related to the axiom of induction.

Theorem 1.5.16 Well-ordering principle. Every non-empty subset of the natural numbers $\mathbb{N}$ has a minimum.

Proof. Let $U \subseteq \mathbb{N}$ be a subset of the natural numbers $\mathbb{N}$, and suppose that U does not have a minimum. We proceed by strong induction on $n \in \mathbb{N}$ to show that $n \notin U$.

Base case. Suppose that $n = 0$, and suppose for a contradiction that $n \in U$. Since $0 \leq n$ for all natural numbers $n \in \mathbb{N}$ by [Lemma 1.3.2 0 is minimal](#) and (2) of [Proposition 1.5.15 Extremal elements are extrema in totally ordered sets](#), then $n = \min(U)$. This contradiction implies that $n \notin U$.

Inductive step. Suppose for the inductive hypothesis that there is some natural number $n \in \mathbb{N}$ so that $k \notin U$ for all natural numbers $k \in \mathbb{N}$ so that $0 \leq k \leq n$. Suppose for a contradiction that $n + 1 \in U$, and let $u \in U$. Totality implies that either $u \leq n$ or $n \leq u$. The inductive hypothesis implies that $u \not\leq n$, so that $n \leq u$.

So $u = n + p$ for some natural number $p \in \mathbb{N}$. If $p = 0$, then

$$n = n + 0 = n + p = u \in U,$$

and so $p \neq 0$. Thus $p = q + 1$ for some natural number $q \in \mathbb{N}$, and so

$$n + 1 \leq n + q + 1 = n + p = u.$$

In summary, we have shown that $n + 1 \leq u$ for all $u \in U$, so that $\min(U) = n + 1$. This contradiction implies that $n + 1 \notin U$.

We conclude that $n \notin U$ for all natural numbers $n \in \mathbb{N}$, and so $U = \emptyset$. ■

The [Well-ordering principle](#) is sometimes taken to be an axiom and sometimes derived from the axiom of induction as in the above proof. In fact, these results are logically equivalent, as one can derive the axiom of induction from the [Well-ordering principle](#).

We will revisit ordering relations in the next chapter, when we extend the standard ordering on the natural numbers to many of the extensions of the natural number system which we construct.

1.5.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

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1.5.5 Reading Questions

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

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1.6 Cardinality and Countability [INCOMPLETE]

In this section, we define a notion of the size of a set called **cardinality**. The cardinality of a set is a generalization of the number of elements of a set which contains only finitely many elements based on the notion of bijections. First, we will develop these notions for finite sets and then extend them to infinite sets. At first glance, it may seem that there is essentially only one kind of infinite set. We will show that this is not the case; in fact, there are infinitely many “sizes” of infinite sets with respect to this notion of cardinality.

1.6.1 Finiteness

Before we brave the unintuitive mess which is the realm of infinite sets, we will rigorously develop a notion of size for sets which contain only finitely many

elements. Like in the infinite case, these finite cardinalities will be based on bijections.

Proposition 1.6.1 Injections of finite sets. *For any natural numbers $m, n \in \mathbb{N}$, there exists an injective map $\{1, \dots, m\} \rightarrow \{1, \dots, n\}$ if and only if $m \leq n$.*

Proof. If $m \leq n$, then $\{1, \dots, m\} \subseteq \{1, \dots, n\}$, and so the inclusion map $\iota_{\{1, \dots, m\}}^{\{1, \dots, n\}}: \{1, \dots, m\} \rightarrow \{1, \dots, n\}$ is an injective map from $\{1, \dots, m\}$ to $\{1, \dots, n\}$. So it suffices to show the reverse implication.

To that end, consider the subset

$$S = \{m \in \mathbb{N} \mid \text{There are } n \in \mathbb{N} \text{ and } f: \{1, \dots, m\} \rightarrow \{1, \dots, n\} \text{ so that } m \not\leq n \text{ and } f \text{ is injective}\}$$

of the natural numbers $\mathbb{N}$. It suffices to show that $S = \emptyset$. To that end, we first note that $0 \leq n$ for all natural numbers $n \in \mathbb{N}$, so that $0 \notin S$.

Now let $m \in \mathbb{N}$ be a natural number, and suppose that $m + 1 \in S$. Then there is a natural number $n \in \mathbb{N}$ so that $m + 1 \not\leq n$ and an injective function $f: \{1, \dots, m + 1\} \rightarrow \{1, \dots, n\}$. We note that $n \neq 0$, since there are no functions from a non-empty set $\{1, \dots, m + 1\}$ to the empty set $\emptyset$. Thus $n = p + 1$ for some natural number $p \in \mathbb{N}$.

If $m \leq p$, then $m + 1 \leq p + 1 = n$, and so $m \not\leq p$. It therefore suffices to construct an injective function $\{1, \dots, m\} \rightarrow \{1, \dots, p\}$. To that end, fix a bijection $g: \{1, \dots, p + 1\} \rightarrow \{1, \dots, p + 1\}$ so that $g(f(m + 1)) = p + 1$, and let $h: \{1, \dots, m\} \rightarrow \{1, \dots, p\}$ be the function defined by the formula

$$h(j) = g(f(j)).$$

Since g and f are injective, h is injective by (a) of [provisional cross-reference: problem-compositions-of-injections-surjections-and-bijections]. We conclude that $n \in S$.

In summary, we have shown that $0 \notin S$ and for all natural numbers $n \in \mathbb{N}$, if $n + 1 \in S$, then $n \in S$. These conclusions are equivalent to $0 \in \mathbb{N} \setminus S$ and for all natural numbers $n \in \mathbb{N}$, if $n \in \mathbb{N} \setminus S$, then $n + 1 \in \mathbb{N} \setminus S$. We conclude that $\mathbb{N} \setminus S = \mathbb{N}$ by the axiom of induction, and so $S = \emptyset$. ■

Corollary 1.6.2 Finite cardinality is well-defined. *For any set X , if there are natural numbers $m, n \in \mathbb{N}$ and bijections $f: \{1, \dots, m\} \rightarrow X$ and $g: \{1, \dots, n\} \rightarrow X$, then $m = n$.*

Proof. Under the given assumptions, $g^{-1} \circ f: \{1, \dots, m\} \rightarrow \{1, \dots, n\}$ and $f^{-1} \circ g: \{1, \dots, n\} \rightarrow \{1, \dots, m\}$ are injective functions by (a) of [provisional cross-reference: problem-compositions-of-injections-surjections-and-bijections].

Proposition 1.6.1 Injections of finite sets now implies that $m \leq n$ and $n \leq m$, and so $m = n$ by (2) of Proposition 1.3.15 Comparison is a total order. ■

Definition 1.6.3 Finiteness; cardinality. A set X is called **finite** if there is a natural number $n \in \mathbb{N}$ and a bijection $\{1, \dots, n\} \rightarrow X$, in which case n is uniquely defined by this property and is called the **cardinality** of X and denoted $|X| = n$. A set X is called **infinite** if no such number and bijection exist. ◆

Proposition 1.6.4 Maps and cardinality. *Let $f: X \rightarrow Y$ be a map from a set X to a set Y .*

1. *If f is injective and Y is finite, then X is finite and $|X| \leq |Y|$.*

2. If f is surjective and X is finite, then Y is finite and $|Y| \leq |X|$.
3. If f is bijective and either X or Y is finite, then both X and Y are finite and $|X| = |Y|$.

Proof. ■

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1.6.2 Cardinal Comparison

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1.6.3 Countability

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1.6.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

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Chapter 2

Constructing the Number Systems

In this section, we provide rigorous constructions of well-known number systems including the **integers** $\mathbb{Z}$, the **rational numbers** $\mathbb{Q}$, the **real numbers** $\mathbb{R}$, and the **complex numbers** $\mathbb{C}$. These number systems, particularly the real and complex number systems, provide the backdrop for much of the discipline of analysis.

2.1 The Integers and the Rational Numbers

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Exercises

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Reading Questions

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2.2 Constructing the Real Numbers

In this section, we construct the **real number system** $\mathbb{R}$ by **Dedekind cuts**, and we explore the resultant properties of arithmetic and comparison. However, we will find the existence of such a formal construction and its properties more useful than the particular details of the construction itself, and we will shortly leave behind the perspective suggested by our construction.

2.2.1 Cutting the Rational Numbers

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Definition 2.2.1 Dedekind cut. A proper, non-empty subset $0 \subseteq X \subseteq \mathbb{Q}$ of the rational numbers $\mathbb{Q}$ is called a **Dedekind cut** if it satisfies the following conditions:

1. For all rational numbers $q_1, q_2 \in \mathbb{Q}$, if $q_1 \leq q_2$ and $q_2 \in X$, then $q_1 \in X$.
2. X does not have a maximum; that is, for all elements $q_1 \in X$, there is an element $q_2 \in X$ so that $q_1 < q_2$.

◆

Example 2.2.2 Dedekind cuts. For each rational number $q \in \mathbb{Q}$, consider the subset

$$X_q = \{p \in \mathbb{Q} \mid p < q\}$$

of the rational numbers $\mathbb{Q}$. We observe that X_q is a proper and non-empty subset of $\mathbb{Q}$, since it contains $q - 1$ but not q itself. Moreover, for all rational numbers $p_1, p_2 \in \mathbb{Q}$, if $p_1 \leq p_2$ and $p_2 \in X_q$, then

$$p_1 \leq p_2 < q,$$

and so $p_1 \in X_q$. Finally, for all elements $p_1 \in X$, we observe that

$$p < p + \frac{q-p}{2} < p + (q-p) = q,$$

so that $p + \frac{q-p}{2} \in X_q$. In summary, we have shown that associated to each rational number $q \in \mathbb{Q}$ is a Dedekind cut X_q . ▲

We will see that the above construction realizes the rational numbers $\mathbb{Q}$ as a subset of the set of Dedekind cuts. The Dedekind cuts which arise in this manner will be called **rational**. We will shortly see that not every Dedekind cut is rational.

Lemma 2.2.3 Rational Dedekind cuts. *A Dedekind cut X is rational if and only if its complement $\mathbb{Q} \setminus X$ has a minimum.*

Proof. Let X be a Dedekind cut, and suppose that $X = X_q$ for some rational

number $q \in \mathbb{Q}$. Note that $X = X_q$ has complement

$$\mathbb{Q} \setminus X = \mathbb{Q} \setminus X_q = \{p \in \mathbb{Q} \mid p \notin X_q\} = \{p \in \mathbb{Q} \mid p \geq q\}.$$

This set contains q , since $q \geq q$. Moreover, $p \geq q$ for all elements $p \in \mathbb{Q} \setminus X_q$. Thus $\min(\mathbb{Q} \setminus X) = q$.

Conversely, now suppose that $\mathbb{Q} \setminus X$ has a minimum element, say $q = \min(\mathbb{Q} \setminus X) \in \mathbb{Q}$. Let $p \in \mathbb{Q}$, and suppose that $p \geq q$. If $p \in X$, then that would imply that $q \in X$. This contradiction implies that $p \notin X$. On the other hand, if $p < q$, then $p \notin \mathbb{Q} \setminus X$, since $\min(\mathbb{Q} \setminus X) = q$. Thus

$$X = \{p \in \mathbb{Q} \mid p < q\} = X_q$$

is rational. ■

Proposition 2.2.4 Irrational Dedekind cuts. *Not all Dedekind cuts are rational.*

Proof. Consider the subset

$$X = \{q \in \mathbb{Q} \mid q < 0 \text{ or } q^2 < 2\}$$

of the rational numbers $\mathbb{Q}$. We observe that X is a non-empty and proper subset of $\mathbb{Q}$, since it contains 1 but not 2. Let $p_1, p_2 \in \mathbb{Q}$ be rational numbers, and suppose that $p_1 \leq p_2$ and $p_2 \in X$. If $p_2 < 0$, then

$$p_1 \leq p_2 < 0,$$

and so $p_1 \in X$. On the other hand, if $p_2^2 < 2$, then either $p_1 < 0$ or $0 \leq p_1 \leq p_2$, in which case $p_1^2 \leq p_2^2 < 2$. In either case, we conclude that $p_1 \in X$.

Finally, let $q \in X$. If $q \leq 0$, then let $p = 1$, and note that $p \in X$ and $q < p$. On the other hand, suppose that $q > 0$ and $q^2 < 2$, and consider the rational number

$$p = \frac{2q + 2}{2 + q} \in \mathbb{Q}.$$

Note that

$$p = \frac{2q + 2}{2 + q} = \frac{2q + q^2}{2 + q} + \frac{2 - q^2}{2 + q} = q + \frac{2 - q^2}{2 + q} > q,$$

and

$$\begin{aligned} 2 - p^2 &= \frac{2(2 + q)^2}{(2 + q)^2} - \frac{(2q + 2)^2}{(2 + q)^2} \\ &= \frac{4 - 2q^2}{(2 + q)^2} = \frac{2(2 - q^2)}{(2 + q)^2} > 0, \end{aligned}$$

so that $p^2 < 2$; that is, $p \in X$. We conclude that X is a Dedekind cut.

We note that X has complement

$$\mathbb{Q} \setminus X = \{q \in \mathbb{Q} \mid q \geq 0 \text{ and } q^2 \geq 2\}.$$

In fact, since $q^2 \neq 2$ for all rational numbers $q \in \mathbb{Q}$,

$$\mathbb{Q} \setminus X = \{q \in \mathbb{Q} \mid q \geq 0 \text{ and } q^2 > 2\}.$$

This set has no minimum element, and so X is not rational. ■

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2.2.2 Arithmetic of Dedekind Cuts

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2.2.3 Comparison of Dedekind Cuts

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2.2.4 Exercises

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2.3 The Complete Ordered Field

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Exercises

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Reading Questions

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2.4 Beyond the Real Number System

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Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

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Reading Questions

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

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Part II

Point-Set Topology

Chapter 3

Metric Spaces

In this chapter, we introduce metric spaces, which generalize the notion of distance in the physical world, and are the natural setting for beginning our study of analysis. Analysis as a field of mathematics is concerned with the limiting behavior of mathematical objects. In a metric space, it makes sense to talk about the distance between any two points. Armed only with this notion of distance, we will define what it means

- for sequences and functions to converge to a limit;
- for sets to be open and/or closed;
- for maps to be continuous; and
- for metric spaces to be complete.

However, in spite of the strength and applicability of the results to be proven in this chapter, we will eventually find that we require further generalizations of the above notions, which will occur in the next chapter.

3.1 Introduction to Metric Geometry

To begin, we introduce a generalization of our usual notion of distance which is called a **metric**. The properties of physical distance make it an extremely useful tool for formalizing the analytical notions mentioned above.

3.1.1 Introduction to Metrics

Not all properties of physical distance are generic enough to generalize to other settings in which we might want to measure distance-like quantities. This question of which properties of a specific example (usually Euclidean space) are salient enough to merit requirement in a generalization is ubiquitous throughout mathematical exposition, and will recur frequently throughout this text.

Definition 3.1.1 Metric space. A real-valued function $\rho: X^2 \rightarrow \mathbb{R}$ on the Cartesian square X^2 of a set X , which is called the **ground set** and whose elements are called **points**, is a **metric** on X if it has the following properties:

1. *Non-degeneracy.*

For any points $x_1, x_2 \in X$, $\rho(x_1, x_2) = 0$ if and only if $x_1 = x_2$.

2. *Symmetry.*

$$\rho(x_1, x_2) = \rho(x_2, x_1) \text{ for any points } x_1, x_2 \in X.$$

3. *Triangle inequality.*

$$\rho(x_1, x_3) \leq \rho(x_1, x_2) + \rho(x_2, x_3) \text{ for any points } x_1, x_2, x_3 \in X.$$

In this case, the ordered pair (X, ρ) is called a **metric space**. ◆

Remark 3.1.2 Implied metrics. We will see that, in general, there are many metrics on a given set, no one of which may truly be considered canonical. In fact, even though some distinct metrics give identical answers to point-set topological concerns such as convergence and continuity, in most cases even a choice one of these classes of metrics is arbitrary in some respects.

However, sometimes, if a metric $\rho: X^2 \rightarrow \mathbb{R}$ on a set X is clear from context, previously specified, or implicitly present, some mathematicians will refer to the metric space (X, ρ) by only the underlying set X . We will try to avoid such ambiguities, and we will attempt to always explicitly specify the metric with which we will be working.

Example 3.1.3 Metric spaces. The following are examples of metrics and metric spaces:

1. Let $n \in \mathbb{N}$ be a natural number. The **Euclidean metric** $d: (\mathbb{R}^n)^2 \rightarrow \mathbb{R}$ on n -dimensional real coordinate space $\mathbb{R}^n$ is the function defined by the formula

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}.$$

Similarly, the **Euclidean metric** $d: (\mathbb{C}^n)^2 \rightarrow \mathbb{R}$ on n -dimensional real coordinate space $\mathbb{C}^n$ is the function defined by the formula

$$d(\mathbf{w}, \mathbf{z}) = \sqrt{\sum_{k=1}^n |w_k - z_k|^2}.$$

As defined above, the Euclidean metrics are metrics on $\mathbb{R}^n$ and $\mathbb{C}^n$, respectively. The metric spaces $(\mathbb{R}^n, d)$ and $(\mathbb{C}^n, d)$ are called n -dimensional **real Euclidean space** and n -dimensional **complex Euclidean space**, respectively. 1- and 2-dimensional real Euclidean space $(\mathbb{R}, d)$ and $(\mathbb{R}^2, d)$ are called the **real line** and the **Euclidean plane**, respectively, and 1-dimensional complex Euclidean space $(\mathbb{C}, d)$ is called the **complex plane**.

2. Consider the real-valued function $\rho: (0, \infty)^2 \rightarrow \mathbb{R}$ defined by the formula

$$\rho(x, y) = \left| \ln \left(\frac{x_1}{x_2} \right) \right|$$

As defined above, ρ is a metric on the open interval $(0, \infty)$, and so $((0, \infty), \rho)$ is a metric space.

3. The **discrete metric** $\delta_X: X^2 \rightarrow \{0, 1\}$ on a set X is the function defined by the formula

$$\delta_X(x_1, x_2) = \begin{cases} 1 & x_1 = x_2 \\ 0 & x_1 \neq x_2. \end{cases}$$

As defined above, the discrete metric δ_X is a metric on X , and the metric space (X, δ_X) is called a **discrete metric space**. ▲

The definition of a metric is pretty abstract, and it may be difficult to see how or why requiring only nondegeneracy, symmetry, and the triangle inequality would give a useful generalization of distance, especially given the strange and outlandish properties of the discrete metric described above, which we will explore later in this section. However, requiring the three above properties actually forces a metric to have many of the properties of our usual notion of distance, and so we can often use our knowledge of distance in the physical world to inform our intuition about metric spaces.

As an example, consider for a moment the notion of negative distance. In the real world, it makes no sense to consider negative distances between physical objects, and we would be equally confused by negative distances between points in a metric space. However, the defining axioms of a metric do not a priori explicitly forbid this nonsensical outcome from occurring. At first glance, it might seem like a metric might be allowed to take negative values, even though the three examples of a metric given above are certainly nonnegative. In fact, this is never the case.

Lemma 3.1.4 Metrics are non-negative. *A metric takes only non-negative values.*

Proof. Let ρ be a metric on a set X , and note that

$$\begin{aligned}\rho(x_1, x_2) &= \frac{2\rho(x_1, x_2)}{2} = \frac{\rho(x_1, x_2) + \rho(x_1, x_2)}{2} \\ &= \frac{\rho(x_1, x_2) + \rho(x_2, x_1)}{2} \geq \frac{\rho(x_1, x_1)}{2} = \frac{0}{2} = 0\end{aligned}$$

for all points $x_1, x_2 \in X$. ■

The above argument shows that, just as in the physical world, the distance between any two points in a metric space must be nonnegative. In fact, unlike many of the spaces which are studied by analysts (some of which we will encounter in the following chapters), metric spaces can be surprisingly accessible using only our intuitions about distance in the physical world. For example, the triangle inequality expresses that there are no “shortcuts” between two points in a metric space through a third point; in geometric terms, we require that no leg of a triangle may exceed in length the sum of the lengths of the other two legs. While the triangle inequality establishes an upper bound on the length of a leg of a triangle, the following property of a metric is called the **reverse triangle inequality**, and it establishes a lower bound on the length of a leg of a triangle.

Proposition 3.1.5 Reverse triangle inequality. *Let (X, ρ) be a metric space. Then*

$$\rho(x_1, x_2) \geq |\rho(x_1, x_3) - \rho(x_2, x_3)|$$

for all points $x_1, x_2, x_3 \in X$.

Proof. If $\rho(x_1, x_3) \leq \rho(x_2, x_3)$, then $\rho(x_1, x_3) - \rho(x_2, x_3) \leq 0$, and so

$$\begin{aligned}|\rho(x_1, x_3) - \rho(x_2, x_3)| &= \rho(x_2, x_3) - \rho(x_1, x_3) \\ &\leq \rho(x_2, x_1) + \rho(x_1, x_3) - \rho(x_1, x_3) = \rho(x_1, x_2).\end{aligned}$$

On the other hand, if $\rho(x_1, x_3) \geq \rho(x_2, x_3)$, then $\rho(x_1, x_3) - \rho(x_2, x_3) \geq 0$, and so

$$\begin{aligned}|\rho(x_1, x_3) - \rho(x_2, x_3)| &= \rho(x_1, x_3) - \rho(x_2, x_3) \\ &\leq \rho(x_1, x_2) + \rho(x_2, x_3) - \rho(x_2, x_3) = \rho(x_1, x_2).\end{aligned}$$

■

We have just seen that metric spaces often agree with our geometric intuitions about distance. However, we must be careful not to assume too much about the geometry of metric spaces in general; even though many of the usual properties of distance in the physical world are implied by the defining axioms of a metric space, there is still lots of room for geometric peculiarities. In particular, some of the metrics defined above disagree with our geometric intuitions in confusing ways. This is because they satisfy a stronger form of the triangle inequality than is true in the physical world. We will explore these **ultrametrics** in the reading questions for this section.

3.1.2 Constructing New Metrics

We will now see how to construct new metrics out of given metric spaces. Given a metric space, there is a natural metric on any subset of its ground set. This restriction of the metric structure to a subset is done in the obvious way; the distance between any points in the subset is just their distance when considered as points in the bigger space. In this way, a metric on a set is said to induce a metric on all of its subsets.

Definition 3.1.6 Induced metric. Let $U \subseteq X$ be a subset of a metric space (X, ρ) . The **induced metric** on U is the restriction $\rho|_{U^2} : U^2 \rightarrow \mathbb{R}$ of ρ to U^2 . The metric space $(U, \rho|_U)$ is called a **metric subspace** of the metric space (X, ρ) .

In practice, we often write that ρ is a metric on U , even if the domain of ρ is strictly larger than U^2 . We will not distinguish between a metric and its restriction to a smaller space. $\blacklozenge$

Example 3.1.7 Discrete metric subspaces. Let $U \subseteq X$ be a subset of a set X . The discrete metric δ_U on U is induced on U by the discrete metric δ_X on X ; that is, $\delta_X|_{U^2} = \delta_U$. Thus the discrete metric space (U, δ_U) is a metric subspace of the discrete metric space (X, δ_X) . $\blacktriangle$

Thus far, we have been very careful when working in a metric space to specify not only the ground set but also the metric. This is because we can in general find many metrics on a given underlying set. In fact, given one or more metrics on a set, there are several ways of constructing new metrics on that set. In particular, any finite positive pointwise linear combination of metrics is itself a metric.

Proposition 3.1.8 Linear combinations of metrics. *Any positive real linear combination of metrics is a metric.*

Proof. Let X be a set, and consider positive real numbers $\alpha_1, \dots, \alpha_n \in (0, \infty]$ and metrics $\rho_1, \dots, \rho_n : X^2 \rightarrow \mathbb{R}$ on X . Denote by $\rho : X^2 \rightarrow \mathbb{R}$ the linear combination defined by the formula

$$\rho(x_1, x_2) = \sum_{k=1}^n \alpha_k \rho_k(x_1, x_2).$$

First let $x_1, x_2 \in X$ be points, and suppose that $\rho(x_1, x_2) = 0$. Then

$$\sum_{k=1}^n \alpha_k \rho_k(x_1, x_2) = \rho(x_1, x_2) = 0.$$

Since $\alpha_k > 0$ and $\rho_k(x_1, x_2) \geq 0$ for all indices $1 \leq k \leq n$, we conclude that $\rho_k(x_1, x_2) = 0$ for all indices $1 \leq k \leq n$. In particular, $x_1 = x_2$.

Conversely, now suppose that $x_1 = x_2$. Then

$$\rho(x_1, x_2) = \sum_{k=1}^n \alpha_k \rho_k(x_1, x_2) = \sum_{k=1}^n \alpha_k \cdot 0 = 0.$$

Thus ρ is non-degenerate.

Note that

$$\rho(x_1, x_2) = \sum_{k=1}^n \alpha_k \rho_k(x_1, x_2) = \sum_{k=1}^n \alpha_k \rho_k(x_2, x_1) = \rho(x_2, x_1)$$

for all points $x_1, x_2 \in X$, and so ρ is symmetric.

Finally, note that

$$\begin{aligned} \rho(x_1, x_3) &= \sum_{k=1}^n \alpha_k \rho_k(x_1, x_3) \leq \sum_{k=1}^n \alpha_k (\rho_k(x_1, x_2) + \rho_k(x_2, x_3)) \\ &= \sum_{k=1}^n \alpha_k \rho_k(x_1, x_2) + \sum_{k=1}^n \alpha_k \rho_k(x_2, x_3) = \rho(x_1, x_2) + \rho(x_2, x_3) \end{aligned}$$

for all points $x_1, x_2, x_3 \in X$, and so ρ satisfies the triangle inequality. Since ρ is non-degenerate, symmetric, and satisfies the triangle inequality, it is a metric on X . ■

In particular, the sum of two metrics is a metric, and any positive real multiple of a metric is a metric.

3.1.3 Normed Vector Spaces

The ground sets of many of the metric spaces which we will investigate over the course of this chapter can also be considered as the ground sets of vector spaces. Moreover, in practically all of these cases, the linear-algebraic structure of the vector space is compatible with the analytical structure of the metric, in a way which we will make precise shortly. We now introduce a mathematical structure called a **normed vector space**, the study of which is called **functional analysis** and lies at the intersection of linear algebra and mathematical analysis. Aptly named, a normed vector space is a vector space equipped with the additional structure of a norm. Just as metrics generalize the notion of the distance between two points, norms on vector spaces generalize the related notion of the length of a vector.

Definition 3.1.9 Normed vector space. A real-valued function $\|\cdot\| : V \rightarrow \mathbb{R}$ on a real or complex vector space V is a **norm** on V if it has the following properties:

1. *Non-degeneracy.*

For any vector $\mathbf{v} \in V$, $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

2. *Absolute homogeneity.*

$\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|$ for any scalar α and vector $\mathbf{v} \in V$.

3. *Sub-additivity.*

$\|\mathbf{v}_1 + \mathbf{v}_2\| \leq \|\mathbf{v}_1\| + \|\mathbf{v}_2\|$ for all vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$.

In this case, the ordered pair $(V, \|\cdot\|)$ is called a **normed vector space**. ♦

Normed vector spaces are ubiquitous throughout mathematics. In fact, many of the vector spaces which you will have encountered in any course in linear algebra have canonical norms associated to them; lots of the vector spaces which are familiar to you have been normed vector spaces all along.

Example 3.1.10 Normed vector spaces. The following are examples of normed vector spaces:

1. Let $n \in \mathbb{N}$ be a natural number, and fix a positive real number $p > 0$. The **p -norm** $\|\cdot\|_p : \mathbb{R}^n \rightarrow \mathbb{R}$ on n -dimensional real coordinate space $\mathbb{R}^n$ is the real-valued function defined by the formula

$$\|\mathbf{x}\|_p = \sqrt[p]{\sum_{k=1}^n x_k^p}.$$

Similarly, the **p -norm** $\|\cdot\|_p : \mathbb{C}^n \rightarrow \mathbb{R}$ on n -dimensional complex coordinate space $\mathbb{C}^n$ is the real-valued function defined by the formula

$$\|\mathbf{z}\|_p = \sqrt[p]{\sum_{k=1}^n |z_k|^p}.$$

If $p \geq 1$, then the p -norms are norms on $\mathbb{R}^n$ and $\mathbb{C}^n$, and so $(\mathbb{R}^n, \|\cdot\|_p)$ and $(\mathbb{C}^n, \|\cdot\|_p)$ are normed vector spaces.

In particular, the 2-norms $\|\cdot\|_2$ on $\mathbb{R}^n$ and $\mathbb{C}^n$ are called the **Euclidean norms** on $\mathbb{R}^n$ and $\mathbb{C}^n$. Because the Euclidean norms are so ubiquitous, we will often drop the subscripts in our notation, so that $\|\cdot\| = \|\cdot\|_2$.

2. Let $n \in \mathbb{N}$ be a natural number. The families of p -norms on n -dimensional real and complex coordinate space $\mathbb{R}^n$ and $\mathbb{C}^n$ can be extended to the case $p = \infty$. The **∞ -norm** $\|\cdot\|_\infty : \mathbb{R}^n \rightarrow \mathbb{R}$ on $\mathbb{R}^n$ is the real-valued function defined by the formula

$$\|\mathbf{x}\|_\infty = \max_{1 \leq k \leq n} |x_k|.$$

Similarly, the **∞ -norm** $\|\cdot\|_\infty : \mathbb{C}^n \rightarrow \mathbb{R}$ on $\mathbb{C}^n$ is the real-valued function defined by the formula

$$\|\mathbf{z}\|_\infty = \max_{1 \leq k \leq n} |z_k|.$$

The ∞ -norms are norms on $\mathbb{R}^n$ and $\mathbb{C}^n$, and so $(\mathbb{R}^n, \|\cdot\|_\infty)$ and $(\mathbb{C}^n, \|\cdot\|_\infty)$ are normed vector spaces.

3. Any Hermitian inner product $\langle \cdot, \cdot \rangle$ on a real or complex vector space V induces a norm $\|\cdot\| : V \rightarrow \mathbb{R}$ on V defined by the formula

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

In fact, for all natural numbers $n \in \mathbb{N}$, the norms produced in this manner by the dot product on n -dimensional real and complex coordinate space $\mathbb{R}^n$ and $\mathbb{C}^n$ are precisely the Euclidean norms. For this reason, the dot product is often called the **Euclidean inner product** on $\mathbb{R}^n$ and $\mathbb{C}^n$.

The above examples show that many of our familiar vector spaces have canonical norms readily available. ▲

Remark 3.1.11 *p -norms on 1-dimensional vector spaces.* Note that

$$\|x\|_p = (|x|^p)^{\frac{1}{p}} = |x| = \max(\{|x|\}) = \|x\|_\infty$$

for all real numbers $x \in \mathbb{R}$, and similarly

$$\|z\|_p = (|z|^p)^{\frac{1}{p}} = |z| = \max(\{|z|\}) = \|z\|_\infty$$

for all complex numbers $z \in \mathbb{C}$; that is, all of the p norms (including the case $p = \infty$) on the real numbers $\mathbb{R}$ and the complex numbers $\mathbb{C}$ agree. Such statements do not hold in higher dimensions, although we will see that the p -norms remain related for topological concerns.

Just as in the case of metrics, we would prefer to consider only non-negative vector lengths. It makes no sense to consider negative lengths of physical vectors, and it is just as difficult to make sense of negative values of a norm in a normed vector space. Thankfully, the defining axioms of a norm prevent this from occurring.

Lemma 3.1.12 *Norms are non-negative.* A norm takes only non-negative values.

Proof. Let $\|\cdot\| : V \rightarrow \mathbb{R}$ be a norm on a real or complex vector space V , and note that

$$\begin{aligned} \|\mathbf{v}\| &= \frac{2\|\mathbf{v}\|}{2} = \frac{\|\mathbf{v}\| + \|\mathbf{v}\|}{2} = \frac{\|\mathbf{v}\| + |-1|\|\mathbf{v}\|}{2} \\ &= \frac{\|\mathbf{v}\| + \|-\mathbf{v}\|}{2} \geq \frac{\|\mathbf{v} - \mathbf{v}\|}{2} = \frac{\|\mathbf{0}\|}{2} = \frac{0}{2} = 0 \end{aligned}$$

for all vectors $\mathbf{v} \in V$. ■

Theorem 3.1.13 *Normed vector spaces are metric spaces.* A norm $\|\cdot\| : V \rightarrow \mathbb{R}$ on a real or complex vector space V induces a metric $\rho : V^2 \rightarrow \mathbb{R}$ defined by the formula

$$\rho(\mathbf{v}_1, \mathbf{v}_2) = \|\mathbf{v}_1 - \mathbf{v}_2\|.$$

Proof. First, let $\mathbf{v}_1, \mathbf{v}_2 \in V$ be vectors, and suppose that $\rho(\mathbf{v}_1, \mathbf{v}_2) = 0$. Then

$$\|\mathbf{v}_1 - \mathbf{v}_2\| = \rho(\mathbf{v}_1, \mathbf{v}_2) = 0,$$

and so $\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$; that is, $\mathbf{v}_1 = \mathbf{v}_2$. Conversely, if $\mathbf{v}_1 = \mathbf{v}_2$, then

$$\rho(\mathbf{v}_1, \mathbf{v}_2) = \|\mathbf{v}_1 - \mathbf{v}_2\| = \|\mathbf{0}\| = 0.$$

Thus ρ is non-degenerate.

Since

$$\begin{aligned} \rho(\mathbf{v}_1, \mathbf{v}_2) &= \|\mathbf{v}_1 - \mathbf{v}_2\| = |-1|\|\mathbf{v}_1 - \mathbf{v}_2\| \\ &= \|\mathbf{v}_2 - \mathbf{v}_1\| = \rho(\mathbf{v}_2, \mathbf{v}_1) \end{aligned}$$

for all vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$, ρ is symmetric.

Finally, since

$$\begin{aligned} \rho(\mathbf{v}_1, \mathbf{v}_3) &= \|\mathbf{v}_1 - \mathbf{v}_3\| = \|(\mathbf{v}_1 - \mathbf{v}_2) + (\mathbf{v}_2 - \mathbf{v}_3)\| \\ &\leq \|\mathbf{v}_1 - \mathbf{v}_2\| + \|\mathbf{v}_2 - \mathbf{v}_3\| = \rho(\mathbf{v}_1, \mathbf{v}_2) + \rho(\mathbf{v}_2, \mathbf{v}_3) \end{aligned}$$

for all vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in V$, ρ satisfies the triangle inequality. Since ρ is non-degenerate, symmetric, and satisfies the triangle inequality, it is a metric on V . ■

So, given a normed vector space, we may easily convert it into a metric space by defining the distance between two vectors to be the norm of their difference. Many important metric spaces arise as a result of this construction.

Example 3.1.14 p -metrics. Let $n \in \mathbb{N}$ be a natural number, and fix an extended real number $1 \leq p \leq \infty$. The metrics d_p on n -dimensional real and complex coordinate spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ by the p -norms $\|\cdot\|_p$ are called **p -metrics**. These p -metrics are defined by the formula

$$d_p(\mathbf{w}, \mathbf{z}) = \|\mathbf{z}_1 - \mathbf{z}_2\|_p = \left(\sum_{k=1}^n |w_k - z_k|^p \right)^{\frac{1}{p}}.$$

In particular, we note that the metrics $d_2 = d$ induced by the Euclidean norms $\|\cdot\|_2 = \|\cdot\|$ are the Euclidean metrics. $\blacktriangle$

So any norm structure on a real or complex vector space can be reanalyzed as also providing a metric structure. However, a metric on a real or complex vector space need not be induced by a norm. Moreover, there exist metric spaces which cannot be constructed from a normed vector space. So, while normed vector spaces are very useful for defining specific metric spaces, they do not capture all of what can be said about metric spaces, and so will not merit any more special consideration just yet. With this illuminating example of normed vector spaces in mind, we now resume our study of metric spaces in their full generality.

In the next section, we introduce one of the most important ideas in the field of analysis: that of convergence to a limit.

3.1.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

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3.1.5 Reading Questions

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

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3.2 Convergence and Limits

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3.3 Limits in the Euclidean Metric

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3.4 Open and Closed Sets

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3.5 Notions of Metric Continuity

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Exercises

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3.6 Completeness and Compactness

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Exercises

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In this chapter, we have seen that a metric allows us to easily develop a notion of closeness for points in a set based on the standard ordering on the nonnegative real numbers, and we have exploited this notion to explore analysis in metric spaces. In the next chapter, we will examine more abstract notions of closeness which will come from objects known as **topologies**.

Chapter 4

Topological Spaces

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4.1 Introduction to Topology

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Exercises

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4.2 Comparison of Topologies

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4.3 Convergence and Separation

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4.4 Continuity in Topological Spaces

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4.5 Connectedness and Compactness

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Part III

Calculus

Chapter 5

Differentiation and Smoothness

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5.1 Differentiability

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Exercises

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5.2 The Mean Value Theorem

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Exercises

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5.3 Applications of the Mean Value Theorem

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5.4 Higher Order Derivatives

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Chapter 6

Riemann Integration

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6.1 Riemann Sums

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6.2 Properties of the Riemann Integral

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6.3 The Fundamental Theorems of Calculus

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6.4 Improper Integration

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Chapter 7

Infinite Series

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7.1 Partial Sums and Infinite Series

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7.2 Tests for Series Convergence

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7.3 Power Series

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7.4 Analytic Functions

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Exercises

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Appendices

Appendix A

Selected Solutions

I · Analytical Foundations

1 · Introductory Set Theory

1.1 · What Is a Set?

1.1.4 · Exercises

1.1.4.1. Power sets. Compute the power set $\mathcal{P}(S)$ of the set $S = \{1, 2, 3\}$.

Solution. Choosing a subset of the set $S = \{1, 2, 3\}$ is equivalent to making a unique sequence of three binary choices of whether to include each element 1, 2, and 3. We see that S has the following 8 subsets: $\emptyset$, $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$, and $\{1, 2, 3\}$. Therefore, the power set $\mathcal{P}(S)$ of the set $S = \{1, 2, 3\}$ is the set

$$\mathcal{P}(S) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

1.1.4.2. Set Operations. Compute the given set operations. Write your answer in roster or set-builder notation, or as a well-known set.

(a) Compute $S \cap T$, where

$$S = \{n \in \mathbb{N} \mid n = 2k \text{ for some } k \in \mathbb{N}\} = \{0, 2, 4, 6, \dots\}$$

is the set of even natural numbers and

$$T = \{n \in \mathbb{N} \mid n = 2k + 1 \text{ for some } k \in \mathbb{N}\} = \{1, 3, 5, 7, \dots\}$$

is the set of odd integers.

Solution. No natural number is both even and odd, and so $S \cap T = \emptyset$.

(b) Compute $S \cup T$, where $S = \{0, 3, 4, 9\}$ and $T = \{1, 3, 6\}$.

Solution. $S \cup T$ contains all the elements of S and all the elements of T , so that $S \cup T = \{0, 1, 3, 4, 6, 9\}$.

(c) Compute $S \setminus T$, where

$$S = \{n \in \mathbb{N} \mid n = 5k \text{ for some } k \in \mathbb{N}\} = \{0, 5, 10, 15, \dots\}$$

is the set of natural numbers divisible by 5, and

$$T = \{n \in \mathbb{N} \mid n = 2k \text{ for some } k \in \mathbb{N}\} = \{0, 2, 4, 6, \dots\}$$

is the set of even natural numbers.

Solution. $S \setminus T$ contains all the odd multiples of 5, so that

$$\begin{aligned} S \setminus T &= \{n \in \mathbb{N} \mid n = 10k + 5 \text{ for some } k \in \mathbb{N}\} \\ &= \{5, 15, 25, 35, \dots\}. \end{aligned}$$

1.1.5 · Reading Questions

1.1.5.1. De Morgan's laws. Let S , T , and U be sets.

(a) Prove that $S \setminus (T \cap U) = (S \setminus T) \cup (S \setminus U)$.

Solution. asdf

(b) Prove that $S \setminus (T \cup U) = (S \setminus T) \cap (S \setminus U)$.

Solution. asdf

Taken together, these results are called **De Morgan's laws**.

1.1.5.2. The distributive properties of set operations. Let S , T , and U be sets.

(a) Prove that $S \cap (T \cup U) = (S \cap T) \cup (S \cap U)$.

Solution. asdf

(b) Prove that $S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$.

Solution. asdf

Taken together, these results are called the **distributive properties** of the set operations.

1.1.5.3. Power sets. Recall that the power set $\mathcal{P}(S)$ of a set S is the set whose elements are all the subsets of S .

(a) Determine whether the following statement is true or false:

The power set of a set contains at least two distinct elements.

If the statement is true, prove it. If the statement is false, disprove it by providing a counterexample.

Solution. asdf

Text after the parts.

1.1.5.4. Set operations and containment. Let S and T be sets.

(a) Prove that $S \cap T = S$ if and only if $S \subseteq T$.

Solution. Since $S \cap T \subseteq S$, it suffices to show that $S \subseteq S \cap T$ if and only if $S \subseteq T$. To that end, note that if $S \subseteq S \cap T$, then (1) of [Proposition 1.1.14 Set operations and containment](#) implies that $S \subseteq T$.

Conversely, we note that $S \subseteq S$, so that if $S \subseteq T$, then (1) of [Proposition 1.1.14 Set operations and containment](#) implies that $S \subseteq S \cap T$.

(b) Prove that $S \cup T = S$ if and only if $T \subseteq S$.

Solution. Since $S \cup S \cup T$, it suffices to show that $S \cup T \subseteq S$ if and only if $T \subseteq S$. To that end, note that if $S \cup T \subseteq S$, then (2) of [Proposition 1.1.14 Set operations and containment](#) implies that $T \subseteq S$.

Conversely, we note that $S \subseteq S$, so that if $T \subseteq S$, then (2) of [Proposition 1.1.14 Set operations and containment](#) implies that $S \cup T \subseteq S$.

(c) Prove that $S \setminus T = S$ if and only if $S \cap T = \emptyset$.

Solution. First suppose that $S \cap T = \emptyset$. Then $x \notin T$ for all $x \in S$, and so

$$S \setminus T = \{x \mid x \in S \text{ and } x \notin T\} = \{x \mid x \in S\} = S.$$

Conversely, now suppose that $S \cap T \neq \emptyset$. Then $x \in S \cap T$, and so $x \in S$ and $x \in T$. Since $x \in T$, $x \notin S \setminus T$. So x is an element of S but not $S \setminus T$; we conclude that $S \setminus T \neq S$.

1.2 · Maps and Functions

1.2.4 · Exercises

1.2.4.1. Images and pre-images. Let $f_4: \{1, 2\} \rightarrow \{\text{red}, \text{blue}\}$ be the map from $\{1, 2\}$ to $\{\text{red}, \text{blue}\}$ defined in (2) of [provisional cross-reference: example-maps]; that is,

$$f_4 = \{(1, \text{blue}), (2, \text{blue})\}.$$

Compute the given images and pre-images. Express your answer in roster notation.

(a) Compute the images $f_4(1)$ and $f_4(2)$.

Answer. $f_4(1) = f_4(2) = \text{blue}$.

Solution. Since $(1, \text{blue}) \in f_4$, the image $f_4(1)$ of 1 under f_4 is

$$f_4(1) = \text{blue}.$$

Similarly, since $(2, \text{blue}) \in f_4$, the image $f_4(2)$ of 2 under f_4 is

$$f_4(2) = \text{blue}.$$

(b) Compute the image $\text{im}(f_4)$.

Answer. $\text{im}(f_4) = \{\text{blue}\}$.

Solution. Since $f_4(x) = \text{blue}$ for all $x \in \{1, 2\}$, the image $\text{im}(f_4)$ of f_4 is

$$\text{im}(f_4) = f_4(\{1, 2\}) = \{\text{blue}\}.$$

(c) Compute the pre-images $f_4^{-1}(\text{red})$ and $f_4^{-1}(\text{blue})$.

Answer. $f_4^{-1}(\text{red}) = \emptyset$ and $f_4^{-1}(\text{blue}) = \{1, 2\}$.

Solution. Since $f_4(x) = \text{blue}$ for all $x \in \{1, 2\}$, the pre-images $f_4^{-1}(\text{red})$ and $f_4^{-1}(\text{blue})$ of red and blue under f_4 are

$$f_4^{-1}(\text{red}) = \{x \in \{1, 2\} \mid f_4(x) = \text{red}\} = \emptyset$$

and

$$f_4^{-1}(\text{blue}) = \{x \in \{1, 2\} \mid f_4(x) = \text{blue}\} = \{1, 2\}.$$

1.2.5 · Reading Questions

1.2.5.1. Composition and “jectivity”. Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be maps between sets X , Y , and Z .

(a) Prove that if f and g are injective, then so is their composition $g \circ f: X \rightarrow Z$.

Solution. Let $x_1, x_2 \in X$, and suppose that $(g \circ f)(x_1) = (g \circ f)(x_2)$. Then

$$g(f(x_1)) = (g \circ f)(x_1) = (g \circ f)(x_2) = g(f(x_2)),$$

and so the injectivity of g implies that $f(x_1) = f(x_2)$. Moreover, the injectivity of f implies that $x_1 = x_2$. In summary, we have shown that for all inputs $x_1, x_2 \in X$, if $(g \circ f)(x_1) = (g \circ f)(x_2)$, then $x_1 = x_2$; that is, $g \circ f$ is injective.

- (b) Prove that if f and g are surjective, then so is their composition $g \circ f: X \rightarrow Z$.

Solution. Let $z \in Z$. Since g is surjective, $z = g(y)$ for some $y \in Y$. Moreover, since f is surjective, $y = f(x)$ for some $x \in X$. Thus

$$z = g(y) = g(f(x)) = (g \circ f)(x).$$

In summary, we have shown that for every output $z \in Z$, there is some input $x \in X$ so that $z = (g \circ f)(x)$; that is, $g \circ f$ is surjective.

- (c) Prove that if f and g are bijective, then so is their composition $g \circ f: X \rightarrow Z$.

Solution. Since f and g are bijective, they are both injective and surjective. (1) and (2) above not imply that their composition $g \circ f$ is both injective and surjective; that is, $g \circ f$ is bijective.

- (d) Prove that if the composition $g \circ f: X \rightarrow Z$ is injective, then so is f .

Solution. Suppose that f is not injective. Then $f(x_1) = f(x_2)$ for some distinct inputs $x_1, x_2 \in X$. Note that

$$(g \circ f)(x_1) = g(f(x_1)) = g(f(x_2)) = (g \circ f)(x_2).$$

Since $(g \circ f)(x_1) = (g \circ f)(x_2)$ but $x_1 \neq x_2$, $g \circ f$ is not injective.

In summary, we have shown that if f is not injective, then $g \circ f$ is not injective. The contrapositive of this implication is the desired result: if $g \circ f$ is injective, then so is f .

- (e) Prove that if the composition $g \circ f: X \rightarrow Z$ is surjective, then so is g .

Solution. Suppose that g is not surjective. Then there is some $z \in Z$ so that $z \neq g(y)$ for all $y \in Y$. Note that

$$(g \circ f)(x) = g(f(x)) \neq z$$

for all $x \in X$, and so $g \circ f$ is not surjective.

In summary, we have shown that if g is not surjective, then $g \circ f$ is not surjective. The contrapositive of this implication is the desired result: if $g \circ f$ is surjective, then so is g .

Appendix B

List of Symbols

Symbol	Description	Page
$x \in S$	x is an element of S	2
$x \notin S$	x is not an element of S	2
$\emptyset$	the empty set	3
$P \wedge Q$	P and Q	5
$P \vee Q$	P or Q	5
$\neg P$	not P	5
$S \cap T$	the intersection of S and T	5
$S \cup T$	the union of S and T	5
$S \setminus T$	the complement of T in S	5
(x, y)	the ordered pair with first entry x and second entry y	10
$X \times Y$	the Cartesian product of X with Y	10
$f: X \rightarrow Y$	a map f from X to Y	11
$f: x \mapsto y$	a map f maps an input x to an output $y = f(x)$	11
ι_U^X	the inclusion map of U in X .	12
id_X	the identity map of a set X	12
$f(x)$	the image of an input x under a map f	12
$f(U)$	the image of a set U of inputs under a map f	12
$\text{im}(f)$	the image of a map f	12
$f^{-1}(V)$	the pre-image of a set V of outputs under a map f	12
$f^{-1}(y)$	the pre-image of an output y under a map f	12
$f _U$	the restriction of a map f to a subset U of its domain	13
$g \circ f$	the composition of a map g with a map f	13
$(x_j)_{j \in \mathcal{J}}$	a family of objects x_j indexed by $j \in \mathcal{J}$	17
$\{x_j\}_{j \in \mathcal{J}}$	a set of objects x_j indexed by $j \in \mathcal{J}$	17
$\bigcap_{j \in \mathcal{J}} U_j$	the intersection of a family $(U_j)_{j \in \mathcal{J}}$ of sets U_j	17
$\bigcup_{j \in \mathcal{J}} U_j$	the union of a family $(U_j)_{j \in \mathcal{J}}$ of sets U_j	17
$\mathbb{N}$	the set of natural numbers	21
succ	the successor function	21
add	the addition operation	21
mult	the multiplication operation	21
comp	the standard ordering	21

(Continued on next page)

Symbol	Description	Page
$\prod_{j \in \mathcal{J}} U_j$	the Cartesian product of a family $(U_j)_{j \in \mathcal{J}}$ of sets U	31
$\pi_{\prod_{j \in \mathcal{J}} U_j}^{U_k}$	the projection map of a Cartesian product $\prod_{j \in \mathcal{J}} U_j$ onto one of its factors U_k	31
X^m	the m th Cartesian power of a set X	33
$\sum_{j=1}^m n_j$	recursive summation notation	34
$\prod_{j=1}^m n_j$	recursive product notation	34
R_f	the pullback of a finitary relation R along a map f	35
$R _U$	the restriction of a finitary relation R to a subset U	??
$[x]_{\equiv}$	the equivalence class of x under $\equiv$	36
$X/\equiv$	the quotient of X by $\equiv$	36
$\max(U)$	the maximum of U	40
$\min(U)$	the minimum of U	40
$ X $	the cardinality of X	44
X_q	the rational Dedekind cut associated to q	??
δ_X	the discrete metric on X	54
$\ \mathbf{v}\ $	The norm of a vector $\mathbf{v}$	57
$\ \cdot\ _p$	the p -norm	58
$\ \cdot\ _p$	the p -norm	58
d_p	the p -metric	??

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Colophon

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